

Curie–Weiss & Diffusion



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September 28, 2026

Road map of the document

These notes study a *mixture-of-Boltzmann* model coupled to an Ornstein–Uhlenbeck diffusion process. The central question is: *at what diffusion time does the posterior over latent patterns concentrate on a single winner?* The exposition is organised in five connected parts.

1. **Static analysis** (Sections 1–2): component factorization, log-partition asymptotics, and static polarization.
2. **Exact diffusion likelihood** (Section 3): closed-form component likelihoods and the scalar evidence $S_\mu(x, t)$.
3. **Selection criteria** (Sections 3.3–4.2): posterior concentration, a geometry-only RMS proxy, and planted recovery via pairwise SNR and a union bound.
4. **Binary templates and score curvature** (Sections 5–6.5): exact binary log-odds, finite- d error, the diffusion score, and exact spinodal times from the Hessian spectrum.
5. **Hierarchical templates** (Section 7): distinct coarse and fine recovery scales, exact spinodals in a balanced two-pair construction, simulation-ready observables, and an exact fixed-time Monte Carlo validation of the predicted recovery scales.

Summary of main results

- **Static mixture (no diffusion).** The component partition functions factorize:

$$Z_\mu = \sum_{a \in \{\pm 1\}^D} e^{\beta \langle h^\mu, a \rangle} = \prod_{i=1}^D 2 \cosh(\beta h_i^\mu) = 2^D e^{DF_\mu^{(D)}}, \quad F_\mu^{(D)} := \frac{1}{D} \sum_{i=1}^D \log \cosh(\beta h_i^\mu).$$

Hence

$$Z = \sum_{\mu=1}^M Z_\mu = 2^D \sum_{\mu=1}^M e^{DF_\mu^{(D)}}, \quad P_0(\mu) = \frac{e^{DF_\mu^{(D)}}}{\sum_\nu e^{DF_\nu^{(D)}}}.$$

If the maximizer of $F_\mu^{(D)}$ has a positive $O(1)$ gap and M is fixed, then $P_0(\mu)$ concentrates exponentially fast on that maximizer.

- **Diffusion likelihood.** Under $x|a, t \sim \mathcal{N}(e^{-t}a, \Delta_t I)$ with $\Delta_t = T(1 - e^{-2t})$,

$$\log P_t(x | \mu) = \log c(x, t) + S_\mu(x, t), \quad S_\mu(x, t) = \sum_{i=1}^D \left[\log \cosh(\beta h_i^\mu + \lambda(t)x_i) - \log \cosh(\beta h_i^\mu) \right],$$

where $\lambda(t) = e^{-t}/\Delta_t$ and $c(x, t)$ is independent of μ .

- **High-noise expansion.** For $\lambda(t)$ small,

$$S_\mu(x, t) = \lambda(t) \langle x, m^\mu \rangle + \frac{\lambda(t)^2}{2} \sum_{i=1}^D x_i^2 (1 - (m_i^\mu)^2) + \dots, \quad m_i^\mu = \tanh(\beta h_i^\mu).$$

- **Two distinct time criteria.** The centred high-noise calculation gives the geometry-only RMS proxy

$$t_{\text{rms}} = \frac{1}{2} \log \left(1 + \frac{\|\Delta m^{\mu\nu}\|_2^2}{L_{\text{rms}}^2 T} \right).$$

Under a planted pattern μ^* , reliable pairwise recovery is instead controlled by

$$\text{SNR}_{\mu^*\nu}(t) \approx \frac{e^{-t}}{\sqrt{\Delta_t}} \frac{\langle m^{\mu^*}, \Delta m^{\mu^*\nu} \rangle}{\|\Delta m^{\mu^*\nu}\|_2}.$$

The two criteria share the same time-dependent factor but answer different questions: score spread versus correct recovery.

- **Binary and spectral results.** Binary templates admit an exact pairwise log-odds formula and an exact finite- d error probability. The Hessian of $\log P_t$ at the balanced origin has mode-dependent spinodals

$$t_{\text{spin},r} = \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2 \gamma_r}{T} \right),$$

where γ_r are eigenvalues of the template kernel.

- **Two-level hierarchy.** For the balanced four-template construction $\{u, v, -u, -v\}$ with $d_H(u, v) = k$, the coarse and fine geometry is carried by eigenvalues $D - k$ and k . This yields distinct recovery and curvature times, both separated asymptotically by $\frac{1}{2} \log((D - k)/k)$ when their respective large-argument conditions hold.
- **Monte Carlo validation.** Exact fixed-time sampling from the balanced two-pair model confirms the separation between coarse and fine recovery. In the detailed benchmark $D = 400$, $k = 20$, $m_0 = 0.9$, and $T = 1$, the empirical 95% crossings are $\hat{t}_{\text{out}} = 2.398$ and $\hat{t}_{\text{in}} = 0.953$, while the corresponding high-noise predictions are 2.371 and 0.972. Parameter sweeps show that the gap shrinks as k approaches $D/2$, and that weak polarization can make the fine 95% target unattainable.

Standing Assumptions

- **Dimension / mixture size.** We consider $D \gg 1$. Unless stated otherwise, M is fixed as $D \rightarrow \infty$ (so $\log M/D \rightarrow 0$ in Laplace/log-sum-exp bounds).
- **Templates are quenched.** The patterns $\{h^\mu\}_{\mu=1}^M$ are treated as fixed; randomness comes from sampling (μ, a, x) in the generative framework.
- **High-noise/Taylor regime.** The $\lambda(t)$ -expansion of $S_\mu(x, t)$ is controlled by $\lambda(t) \max_i |x_i|$. Under the asymptotic $x \approx \mathcal{N}(0, \Delta_t I)$ one typically has $\max_i |x_i| \sim \sqrt{\Delta_t \log D}$, so a convenient sufficient condition is

$$\lambda(t) \sqrt{\Delta_t \log D} \ll 1.$$

- **Two regimes to keep distinct.**
 - (i) *Thermodynamic dominance:* $P_0(\mu)$ already concentrates exponentially on a single μ^* (unique maximizer of $F_\mu^{(D)}$ with a positive gap).
 - (ii) *Balanced mixtures:* several $F_\mu^{(D)}$ are comparable, and diffusion evidence $S_\mu(x, t)$ controls selection.

1 Model, thermodynamics, and diffusion adaptation

We begin by specifying the full generative framework. The model has three layers: (i) a discrete latent pattern μ , (ii) a binary spin configuration a drawn conditionally on μ , and (iii) a continuous noisy observation x produced by an OU diffusion channel. All subsequent analysis—both the static thermodynamics and the dynamic selection theory—follows from this single generative specification.

Generative Framework

1. Sample a pattern index μ from the induced prior

$$P_0(\mu) = \frac{Z_\mu}{\sum_{\nu=1}^M Z_\nu}, \quad Z_\mu = \sum_{a \in \{\pm 1\}^D} e^{\beta \langle h^\mu, a \rangle} = \prod_{i=1}^D 2 \cosh(\beta h_i^\mu).$$

2. Given μ , sample a clean spin configuration $a \in \{\pm 1\}^D$ from the product measure

$$P_0(a \mid \mu) = \prod_{i=1}^D \frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)}, \quad \mathbb{E}[a_i \mid \mu] = \tanh(\beta h_i^\mu).$$

3. Given diffusion time $t \geq 0$, generate an observation via the OU forward channel

$$x \mid a, t \sim \mathcal{N}(e^{-t} a, \Delta_t I), \quad \Delta_t = T(1 - e^{-2t}).$$

4. Infer μ from (x, t) using Bayes:

$$P(\mu \mid x, t) \propto P_0(\mu) P_t(x \mid \mu).$$

1.1 Configuration space and templates

Let $D \gg 1$ and let the configuration (spin) space be the hypercube

$$\mathcal{S}_D := \{\pm 1\}^D, \quad s = (s_i)_{i=1}^D \in \mathcal{S}_D.$$

Fix $M \in \mathbb{N}$ templates (“patterns”)

$$h^\mu = (h_i^\mu)_{i=1}^D \in \mathbb{R}^D, \quad \mu = 1, \dots, M.$$

Throughout, $\beta > 0$ is the inverse temperature. We assign to each spin configuration the probability law

$$P_0(s) \propto \sum_{\mu=1}^M \exp(\beta \langle h^\mu, s \rangle). \quad (1)$$

With the model in hand, we now turn to its static properties—that is, the structure of $P_0(s)$ and $P_0(\mu)$ before any diffusion is introduced.

2 Static Analysis: Thermodynamics of Mixture-of-Boltzmann

The goal of this section is to understand the prior $P_0(\mu)$ over patterns and, in particular, the conditions under which it concentrates on a single dominant pattern. This concentration phenomenon is the static analogue of the “selection” that the diffusion process will later produce dynamically. For each pattern μ , define a *pattern energy*

$$E_\mu(s) := -\langle h^\mu, s \rangle = -\sum_{i=1}^D h_i^\mu s_i.$$

The model is interpretable as a *mixture of Boltzmann factors* (cf. (1)):

$$P_0(s) \propto \sum_{\mu=1}^M \exp(\beta \langle h^\mu, s \rangle).$$

By introducing an auxiliary variable $\mu \in \{1, \dots, M\}$ and defining the joint law

$$P_0(s, \mu) \propto \exp(\beta \langle h^\mu, s \rangle).$$

Then the conditional and marginal are

$$P_0(s \mid \mu) = \frac{\exp(\beta \langle h^\mu, s \rangle)}{Z_\mu} = \prod_{i=1}^D \frac{e^{\beta h_i^\mu s_i}}{2 \cosh(\beta h_i^\mu)}, \quad (2)$$

$$Z_\mu := \sum_{s \in \{\pm 1\}^D} \exp(\beta \langle h^\mu, s \rangle) = \prod_{i=1}^D 2 \cosh(\beta h_i^\mu), \quad (3)$$

$$P_0(\mu) = \frac{Z_\mu}{\sum_{\nu=1}^M Z_\nu}. \quad (4)$$

The full partition function is

$$Z := \sum_{s \in \mathcal{S}_D} \sum_{\mu=1}^M e^{\beta \langle h^\mu, s \rangle} = \sum_{\mu=1}^M Z_\mu = \sum_{\mu=1}^M \prod_{i=1}^D 2 \cosh(\beta h_i^\mu). \quad (5)$$

2.1 Exact factorization and Laplace principle (static dominance)

The partition function Z is a sum of M terms, each of which is a product over D factors. We now rewrite this in a way that makes the large- D asymptotics transparent. Define

$$F_\mu^{(D)} := \frac{1}{D} \sum_{i=1}^D \log \cosh(\beta h_i^\mu). \quad (6)$$

Then

$$\sum_{i=1}^D \log \cosh(\beta h_i^\mu) = D F_\mu^{(D)}.$$

Exponentiating both sides and using $e^{\sum_i \log x_i} = \prod_i x_i$ gives

$$\exp\left(D F_\mu^{(D)}\right) = \exp\left(\sum_{i=1}^D \log \cosh(\beta h_i^\mu)\right) = \prod_{i=1}^D \cosh(\beta h_i^\mu).$$

Therefore, recalling (5) (i.e. $Z = \sum_{\mu=1}^M \prod_{i=1}^D 2 \cosh(\beta h_i^\mu)$), we obtain

$$Z = \sum_{\mu=1}^M \prod_{i=1}^D 2 \cosh(\beta h_i^\mu) = 2^D \sum_{\mu=1}^M \prod_{i=1}^D \cosh(\beta h_i^\mu) = 2^D \sum_{\mu=1}^M \exp\left(D F_\mu^{(D)}\right),$$

i.e.

$$Z = 2^D \sum_{\mu=1}^M \exp\left(D F_\mu^{(D)}\right). \quad (7)$$

This is a sum of exponentials with large parameter D ; its asymptotics are controlled by the standard log-sum-exp bound. For $a_\mu := D F_\mu^{(D)}$, the log-sum-exp bound gives

$$\max_{\mu} a_\mu \leq \log \sum_{\mu=1}^M e^{a_\mu} \leq \max_{\mu} a_\mu + \log M,$$

hence

$$\log 2 + \max_{\mu} F_\mu^{(D)} \leq \frac{1}{D} \log Z \leq \log 2 + \max_{\mu} F_\mu^{(D)} + \frac{\log M}{D}. \quad (8)$$

For fixed M , the bound above is equivalently

$$\frac{1}{D} \log Z = \log 2 + \max_{1 \leq \mu \leq M} F_\mu^{(D)} + O\left(\frac{\log M}{D}\right). \quad (9)$$

If, in addition, each $F_\mu^{(D)}$ converges to a limit F_μ , then

$$\lim_{D \rightarrow \infty} \frac{1}{D} \log Z = \log 2 + \max_{1 \leq \mu \leq M} F_\mu.$$

Strictly speaking, $D^{-1} \log Z$ is the log-partition density (or pressure). The thermodynamic free-energy density is $-(\beta D)^{-1} \log Z$.

Interpretation

The log-partition density, $\frac{1}{D} \log Z$, is determined in the large- D limit by the single pattern μ that maximizes the free-energy density $F_\mu^{(D)}$. All other patterns contribute exponentially subleading corrections. This is the *discrete Laplace principle* in action.

Static posterior over patterns and Laplace dominance. Recall that the mixture-of-Boltzmann prior

$$P_0(s) \propto \sum_{\mu=1}^M e^{\beta \langle h^\mu, s \rangle}$$

admits the exact augmentation $P_0(s, \mu) \propto e^{\beta \langle h^\mu, s \rangle}$, so that

$$P_0(\mu) = \frac{Z_\mu}{\sum_{\nu=1}^M Z_\nu}, \quad Z_\mu = \sum_{s \in \{\pm 1\}^D} e^{\beta \langle h^\mu, s \rangle} = 2^D e^{D F_\mu^{(D)}}, \quad F_\mu^{(D)} = \frac{1}{D} \sum_{i=1}^D \log \cosh(\beta h_i^\mu).$$

Therefore the posterior weight of pattern μ is

$$P_0(\mu) = \frac{e^{D F_\mu^{(D)}}}{\sum_{\nu=1}^M e^{D F_\nu^{(D)}}}. \quad (10)$$

Gap representation and concentration. Let $\mu^* \in \arg \max_{\mu} F_{\mu}^{(D)}$. Factoring out the dominant exponential yields the exact identity

$$Z = 2^D \sum_{\mu=1}^M e^{DF_{\mu}^{(D)}} = 2^D e^{DF_{\mu^*}^{(D)}} \left(1 + \sum_{\nu \neq \mu^*} e^{-D\Delta_{\nu}^{(D)}} \right), \quad \Delta_{\nu}^{(D)} := F_{\mu^*}^{(D)} - F_{\nu}^{(D)} \geq 0. \quad (11)$$

The same factorization applies to the pattern posterior:

$$P_0(\mu^*) = \frac{1}{1 + \sum_{\nu \neq \mu^*} e^{-D\Delta_{\nu}^{(D)}}}, \quad P_0(\nu) = \frac{e^{-D\Delta_{\nu}^{(D)}}}{1 + \sum_{\kappa \neq \mu^*} e^{-D\Delta_{\kappa}^{(D)}}} \quad (\nu \neq \mu^*).$$

Consequently, if the maximizer is unique and there exists $\delta > 0$ such that $\inf_{\nu \neq \mu^*} \Delta_{\nu}^{(D)} \geq \delta$ for all large D , then

$$P_0(\mu^*) \geq \frac{1}{1 + (M-1)e^{-D\delta}} \rightarrow 1,$$

i.e. the posterior concentrates exponentially fast on the dominant pattern. Equivalently, we may say the only relevant contribution in the partition function is given by the pattern maximising $F_{\mu^*}^{(D)}$. Indeed this is an application of the **discrete Laplace principle**: in a finite sum of exponentials with large parameter D , only the largest exponent contributes at leading order, and the gaps $\Delta_{\nu}^{(D)}$ control the rate of concentration.

2.2 Static polarization / alignment

Having established when $P_0(\mu)$ concentrates, we now ask what this implies for the spin configuration s itself. From (2), spins are independent conditional on μ and

$$\mathbb{E}[s_i | \mu] = \tanh(\beta h_i^{\mu}), \quad \mathbb{E}[s | \mu] = (\tanh(\beta h_1^{\mu}), \dots, \tanh(\beta h_D^{\mu})). \quad (12)$$

Thus, if $P_0(\mu)$ concentrates on a single μ^* , then the unconditional law of s approaches $P_0(s | \mu^*)$. Equation (12) identifies its coordinatewise polarization. A scalar concentration statement requires an explicit overlap, as in the binary case below.

Special case: binary templates. If $h_i^{\mu} = H \sigma_i^{\mu}$ with $\sigma_i^{\mu} \in \{\pm 1\}$, then

$$Z_{\mu} = (2 \cosh(\beta H))^D \quad \text{and hence} \quad P_0(\mu) = \frac{1}{M}.$$

Let $m_0 := \tanh(\beta H)$ and define the overlap

$$q_{\mu}(s) := \frac{1}{D} \sum_{i=1}^D \sigma_i^{\mu} s_i.$$

Conditional on μ , the variables $\sigma_i^{\mu} s_i$ are independent, with mean m_0 and variance $1 - m_0^2$. Therefore

$$\mathbb{E}[q_{\mu}(s) | \mu] = m_0, \quad \text{Var}(q_{\mu}(s) | \mu) = \frac{1 - m_0^2}{D}, \quad q_{\mu}(s) \xrightarrow[D \rightarrow \infty]{\mathbb{P}} m_0. \quad (13)$$

Thus the phrase “polarization toward a template” has a precise meaning: a sample has an order-one overlap m_0 with the component that generated it. Binary templates of common amplitude are also exactly in the balanced static regime, so any later preference among their indices is created by the observed noisy sample rather than by unequal partition functions. This completes the static picture. We now introduce the diffusion process that generates noisy observations of a , and study how the posterior $P(\mu | x, t)$ evolves as the noise level changes.

3 Dynamic Analysis: Diffusion Process

We follow the standard OU forward diffusion setting. Let $a \in \{\pm 1\}^D$ denote the latent “clean” configuration (with prior $P_0(a)$ given by (1)). For $t \geq 0$, define the forward trajectory as

$$x \mid a, t \sim \mathcal{N}(e^{-t}a, \Delta_t I), \quad \Delta_t := T(1 - e^{-2t}), \quad (14)$$

where $x = x(t) \in \mathbb{R}^D$ and $T > 0$ is the OU temperature.

Direction of time

Throughout, t is the *forward* diffusion time: $t = 0$ means no noise ($x \approx a$), and $t \rightarrow \infty$ means pure noise ($x \rightarrow \mathcal{N}(0, TI)$). Selection happens during the *reverse* process, i.e. as t decreases. A “large t_{sel} ” means selection occurs early in the reverse diffusion (little denoising needed), while a “small t_{sel} ” means selection requires significant denoising.

3.1 Posterior over patterns under diffusion (expanded notation)

We study the posterior over patterns $\mu \in \{1, \dots, M\}$ given the noised observation x at time t :

$$P(\mu \mid x, t) = \frac{P_0(\mu) P_t(x \mid \mu)}{\sum_{\nu=1}^M P_0(\nu) P_t(x \mid \nu)}. \quad (15)$$

Here

$$P_t(x \mid \mu) = \sum_{a \in \{\pm 1\}^D} P_0(a \mid \mu) p(x \mid a, t), \quad (16)$$

where $p(x \mid a, t) \sim \mathcal{N}(e^{-t}a, \Delta_t I)$. As derived in the Static setting:

$$P_0(a \mid \mu) = \frac{e^{\beta \langle h^\mu, a \rangle}}{Z_\mu}, \quad Z_\mu = \sum_{a \in \{\pm 1\}^D} e^{\beta \langle h^\mu, a \rangle} = \prod_{i=1}^D 2 \cosh(\beta h_i^\mu), \quad (17)$$

The forward diffusion channel is the OU Gaussian kernel

$$x \mid a, t \sim \mathcal{N}(e^{-t}a, \Delta_t I), \quad \Delta_t = T(1 - e^{-2t}), \quad (18)$$

i.e.

$$p(x \mid a, t) = (2\pi\Delta_t)^{-D/2} \exp\left(-\frac{\|x - e^{-t}a\|^2}{2\Delta_t}\right). \quad (19)$$

The next step is to compute $P_t(x \mid \mu)$ in closed form. This is where the product structure of $P_0(a \mid \mu)$ pays off: because spins are independent conditional on μ , the sum over $a \in \{\pm 1\}^D$ factorizes into a product of one-dimensional sums.

$$P_t(x \mid \mu) = \sum_{a \in \{\pm 1\}^D} P_0(a \mid \mu) p(x \mid a, t),$$

plug (17) and (19):

$$P_t(x \mid \mu) = \sum_{a \in \{\pm 1\}^D} \prod_{i=1}^D \left[\frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)} \cdot (2\pi\Delta_t)^{-1/2} \exp\left(-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right) \right]. \quad (20)$$

Swapping summation and product leads to the simplification:

$$P_t(x \mid \mu) = (2\pi\Delta_t)^{-D/2} \prod_{i=1}^D \left[\sum_{a_i=\pm 1} \frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)} \exp\left(-\frac{(x_i - e^{-t}a_i)^2}{2\Delta_t}\right) \right]. \quad (21)$$

Taking log of (21) gives

$$\log P_t(x \mid \mu) = \underbrace{-\frac{D}{2} \log(2\pi\Delta_t)}_{\log c_1(t)} + \sum_{i=1}^D \log \left[\sum_{a_i=\pm 1} \frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)} \exp\left(-\frac{(x_i - e^{-t} a_i)^2}{2\Delta_t}\right) \right]. \quad (22)$$

We now simplify the expression inside the logarithm. Expand

$$(x_i - e^{-t} a_i)^2 = x_i^2 - 2x_i e^{-t} a_i + e^{-2t}.$$

Hence, inside the log in (22),

$$\begin{aligned} & \sum_{a_i=\pm 1} \frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)} \exp\left(-\frac{(x_i - e^{-t} a_i)^2}{2\Delta_t}\right) \\ &= \exp\left(-\frac{x_i^2}{2\Delta_t} - \frac{e^{-2t}}{2\Delta_t}\right) \frac{1}{2 \cosh(\beta h_i^\mu)} \sum_{a_i=\pm 1} \exp\left(a_i \left[\beta h_i^\mu + \frac{e^{-t}}{\Delta_t} x_i\right]\right). \end{aligned} \quad (23)$$

We further simplify by denoting:

$$\lambda(t) := \frac{e^{-t}}{\Delta_t}.$$

Then the summation is just a cosh:

$$\sum_{a=\pm 1} e^{ay} = e^y + e^{-y} = 2 \cosh(y), \quad \text{with } y = \beta h_i^\mu + \lambda(t)x_i.$$

Plugging into (23) yields the exact simplification

$$\sum_{a_i=\pm 1} \frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)} \exp\left(-\frac{(x_i - e^{-t} a_i)^2}{2\Delta_t}\right) = \exp\left(-\frac{x_i^2}{2\Delta_t} - \frac{e^{-2t}}{2\Delta_t}\right) \frac{\cosh(\beta h_i^\mu + \lambda(t)x_i)}{\cosh(\beta h_i^\mu)}. \quad (24)$$

From (21) and (24),

$$P_t(x \mid \mu) = (2\pi\Delta_t)^{-D/2} \exp\left(-\frac{\|x\|^2}{2\Delta_t} - \frac{De^{-2t}}{2\Delta_t}\right) \prod_{i=1}^D \frac{\cosh(\beta h_i^\mu + \lambda(t)x_i)}{\cosh(\beta h_i^\mu)}. \quad (25)$$

Key result: closed-form log-likelihood

We are interested in our log probability, so we read:

$$\log P_t(x \mid \mu) = \underbrace{-\frac{D}{2} \log(2\pi\Delta_t) - \frac{\|x\|^2}{2\Delta_t} - \frac{De^{-2t}}{2\Delta_t}}_{\log c(x,t) \text{ (independent of } \mu)} + \sum_{i=1}^D \left[\log \cosh(\beta h_i^\mu + \lambda(t)x_i) - \log \cosh(\beta h_i^\mu) \right]. \quad (26)$$

Where we isolate the last term as the only μ dependent one. The entire pattern dependence of the log-likelihood is captured by an additive sum of D coordinate contributions. Conditional on a fixed component, the corresponding random coordinates are independent; this is the factorization used in the later moment calculations.

3.2 High-noise Taylor expansion

The exact log-likelihood (26) is useful but hard to interpret directly. In this subsection we expand it in the high-noise regime $t \gg 1$, which reveals the leading mechanism behind pattern selection. Denote the scalar μ -dependent log-evidence by

$$S_\mu(x, t) = \sum_{i=1}^D \left[\log \cosh(\beta h_i^\mu + \lambda(t)x_i) - \log \cosh(\beta h_i^\mu) \right], \quad \lambda(t) := \frac{e^{-t}}{\Delta_t}. \quad (27)$$

In the high-noise regime, for $t \gg 1$, $\lambda(t)$ is small (indeed $\Delta_t = T(1 - e^{-2t}) \rightarrow T$ and hence $\lambda(t) \sim e^{-t}/T$). We expand each summand in powers of $\lambda(t)$.

Step 1: Taylor expansion at one site. Let $f(u) = \log \cosh(u)$. For any u and small δ ,

$$f(u + \delta) = f(u) + \delta f'(u) + \frac{\delta^2}{2} f''(u) + O(\delta^3),$$

$$f(u + \delta) - f(u) = \delta f'(u) + \frac{\delta^2}{2} f''(u) + O(\delta^3),$$

where

$$f'(u) = \tanh(u), \quad f''(u) = \operatorname{sech}^2(u) = 1 - \tanh^2(u).$$

Fix i , set

$$u_i^\mu := \beta h_i^\mu, \quad \delta_i(t) := \lambda(t) x_i.$$

Then

$$\begin{aligned} \log \cosh(u_i^\mu + \delta_i(t)) - \log \cosh(u_i^\mu) &= \delta_i(t) \tanh(u_i^\mu) + \frac{\delta_i(t)^2}{2} \operatorname{sech}^2(u_i^\mu) + O(|\delta_i(t)|^3) \\ &= \lambda(t) x_i \tanh(\beta h_i^\mu) + \frac{\lambda(t)^2 x_i^2}{2} (1 - \tanh^2(\beta h_i^\mu)) + O(\lambda(t)^3 |x_i|^3). \end{aligned} \quad (28)$$

Step 2: Introduce the magnetization vector. Introduce the conditional mean-spin component

$$m_i^\mu := \tanh(\beta h_i^\mu). \quad (29)$$

Then $1 - \tanh^2(\beta h_i^\mu) = 1 - (m_i^\mu)^2$ and (28) becomes

$$\log \cosh(\beta h_i^\mu + \lambda(t) x_i) - \log \cosh(\beta h_i^\mu) = \lambda(t) x_i m_i^\mu + \frac{\lambda(t)^2}{2} x_i^2 (1 - (m_i^\mu)^2) + O(\lambda(t)^3 |x_i|^3).$$

Step 3: General formula over coordinates: $S_\mu(x, t)$. Summing the above identity over $i = 1, \dots, D$ gives

$$S_\mu(x, t) = \lambda(t) \sum_{i=1}^D x_i m_i^\mu + \frac{\lambda(t)^2}{2} \sum_{i=1}^D x_i^2 (1 - (m_i^\mu)^2) + O\left(\lambda(t)^3 \sum_{i=1}^D |x_i|^3\right). \quad (30)$$

(Here the big- O term is obtained by summing the per-site remainders in (28); in particular it is controlled whenever $\lambda(t) \max_i |x_i|$ is small.)

Is the Taylor Expansion accurate? The coordinate-wise remainder in (28) is controlled by $|\delta_i(t)| = \lambda(t) |x_i|$. In particular, a sufficient condition for the expansion to be uniformly accurate is

$$\lambda(t) \max_{1 \leq i \leq D} |x_i| \ll 1.$$

Indeed, assuming $x \approx \mathcal{N}(0, \Delta_t I)$, one typically has $\max_i |x_i| \sim \sqrt{\Delta_t \log D}$ (actually $\sim \sqrt{2\Delta_t \log D}$, but we ignore constants) hence a convenient sufficient condition is

$$\lambda(t) \sqrt{\Delta_t \log D} \ll 1. \quad (31)$$

For $t \gg 1$, $\Delta_t \approx T$ and $\lambda(t) \sim e^{-t}/T$, so (31) holds when $e^{-t} \sqrt{\log D/T} \ll 1$. Whether it remains valid at a proposed selection time must be checked rather than assumed. For example, in the binary model with Hamming distance d , the high-noise SNR criterion gives $e^{-t}/\sqrt{\Delta_t} \asymp L/(m_0 \sqrt{d})$, and therefore

$$\lambda(t) \sqrt{\Delta_t \log D} \asymp \frac{L \sqrt{\log D}}{m_0 \sqrt{d}}.$$

Thus the coordinatewise Taylor expansion is self-consistent when $d \gg \log D$ (up to fixed constants), but it may fail for very close patterns. The exact binary calculation below is included precisely to cover that regime.

Step 4: Leading contribution to posterior odds. Our Object of interest is now:

$$\ell_{\mu\nu}(x, t) = \log \frac{P(\mu | x, t)}{P(\nu | x, t)} = \log \frac{P(\mu)}{P(\nu)} + (S_\mu(x, t) - S_\nu(x, t)).$$

As per our Static analysis, we are able to analyze convergence in probability behaviour by looking at probability ratios.

Keeping only the leading $\lambda(t)$ term in (30) yields

$$\log \frac{P(\mu | x, t)}{P(\nu | x, t)} \approx \log \frac{P(\mu)}{P(\nu)} + \lambda(t) \sum_{i=1}^D x_i (m_i^\mu - m_i^\nu) = \log \frac{P(\mu)}{P(\nu)} + \lambda(t) \langle x, m^\mu - m^\nu \rangle, \quad (32)$$

with $m^\mu = (m_i^\mu)_{i=1}^D$.

3.3 Selection statistics: concentration versus recovery

The exact posterior can be written as

$$P(\mu | x, t) = \frac{P_0(\mu) e^{S_\mu(x, t)}}{\sum_{\nu=1}^M P_0(\nu) e^{S_\nu(x, t)}}.$$

Because this is a softmax, only differences of component scores affect selection. For two patterns μ and ν , define the exact log-posterior odds

$$\ell_{\mu\nu}(x, t) := \log \frac{P(\mu | x, t)}{P(\nu | x, t)} = \log \frac{P_0(\mu)}{P_0(\nu)} + S_\mu(x, t) - S_\nu(x, t). \quad (33)$$

Thus $\ell_{\mu\nu} > 0$ means that μ is more probable than ν , whereas $\ell_{\mu\nu} = c$ means that its posterior odds are larger by the factor e^c .

Posterior concentration around its winner. Let

$$\hat{\mu}(x, t) \in \arg \max_{1 \leq \mu \leq M} P(\mu | x, t)$$

and define the posterior margin

$$\kappa_{\text{post}}(x, t) := \min_{\nu \neq \hat{\mu}(x, t)} \ell_{\hat{\mu}\nu}(x, t). \quad (34)$$

This quantity measures the gap between the winner and its strongest competitor. The exact softmax identity gives

$$P(\hat{\mu} | x, t) = \frac{1}{1 + \sum_{\nu \neq \hat{\mu}} e^{-\ell_{\hat{\mu}\nu}(x, t)}} \geq \frac{1}{1 + (M-1)e^{-\kappa_{\text{post}}(x, t)}}. \quad (35)$$

Consequently, the exact sufficient threshold for $P(\hat{\mu} | x, t) \geq 1 - \varepsilon$ is

$$\kappa_{\text{post}}(x, t) \geq c_{\varepsilon, M} := \log \left(\frac{(M-1)(1-\varepsilon)}{\varepsilon} \right). \quad (36)$$

The slightly stronger but simpler condition $\kappa_{\text{post}} \geq \log((M-1)/\varepsilon)$ is also sufficient. This is a statement about *confidence*: it says that the posterior is concentrated around some winner, but it does not yet say that the winner is the pattern that generated the sample.

4 High-noise selection theory: a proxy and planted recovery

The previous subsection supplied an exact concentration criterion. We now ask at what time it is met. Two related, but logically distinct, quantities will be used:

1. a centred high-noise RMS calculation, which gives a convenient geometry-only *discriminability proxy*;
2. a planted-measure SNR calculation, which gives the probability of recovering the true component.

Keeping these two notions separate prevents the fluctuation scale of a centred random variable from being mistaken for a recovery guarantee.

4.1 Leading-order log-odds and the RMS proxy

At large t , $\lambda(t) = e^{-t}/\Delta_t$ is small. Subtracting the Taylor expansions of S_μ and S_ν gives, at first order,

$$\ell_{\mu\nu}(x, t) \approx \log \frac{P_0(\mu)}{P_0(\nu)} + \lambda(t) \langle x, \Delta m^{\mu\nu} \rangle, \quad \Delta m^{\mu\nu} := m^\mu - m^\nu. \quad (37)$$

The omitted second-order difference is

$$\frac{\lambda(t)^2}{2} \sum_{i=1}^D x_i^2 ((m_i^\nu)^2 - (m_i^\mu)^2),$$

which vanishes identically for equal-amplitude binary templates but not for general templates.

Balanced and statically dominated regimes. If $\log(P_0(\mu)/P_0(\nu)) = \Theta(D)$, the static prior already fixes the winner and there is no non-trivial diffusion-induced selection window. We therefore use the high-noise calculation below only for components with comparable prior weights, typically $\log(P_0(\mu)/P_0(\nu)) = O(1)$. Under the centred high-noise approximation

$$x \approx \mathcal{N}(0, \Delta_t I), \quad (38)$$

we have

$$\langle x, \Delta m^{\mu\nu} \rangle \sim \mathcal{N}(0, \Delta_t \|\Delta m^{\mu\nu}\|_2^2). \quad (39)$$

Hence

$$\text{Var}(\lambda(t) \langle x, \Delta m^{\mu\nu} \rangle) = \frac{e^{-2t}}{\Delta_t} \|\Delta m^{\mu\nu}\|_2^2, \quad (40)$$

and its RMS magnitude is

$$\text{RMS}_0[\lambda(t) \langle x, \Delta m^{\mu\nu} \rangle] = \frac{e^{-t}}{\sqrt{\Delta_t}} \|\Delta m^{\mu\nu}\|_2. \quad (41)$$

The subscript 0 emphasizes that this is computed under the centred null approximation. It measures the spread of the component scores, not the probability of recovering a particular planted pattern.

Geometry-only crossover time. For an order-one RMS level $L_{\text{rms}} > 0$, define the proxy crossover by

$$\frac{e^{-t}}{\sqrt{\Delta_t}} \|\Delta m^{\mu\nu}\|_2 = L_{\text{rms}}. \quad (42)$$

Using $\Delta_t = T(1 - e^{-2t})$ and $u = e^{-2t}$ gives

$$\frac{u \|\Delta m^{\mu\nu}\|_2^2}{T(1 - u)} = L_{\text{rms}}^2,$$

so that

$$t_{\text{rms}}(L_{\text{rms}}; \mu, \nu) = \frac{1}{2} \log \left(1 + \frac{\|\Delta m^{\mu\nu}\|_2^2}{L_{\text{rms}}^2 T} \right). \quad (43)$$

When $\|\Delta m^{\mu\nu}\|_2^2 \gg L_{\text{rms}}^2 T$,

$$t_{\text{rms}} = \frac{1}{2} \log \left(\frac{\|\Delta m^{\mu\nu}\|_2^2}{T} \right) - \log L_{\text{rms}} + o(1). \quad (44)$$

High-dimensional scaling. Assume that, for a given competing pair,

Assumption 1 (Pattern separation). *There exists $c_{\mu\nu} > 0$ such that*

$$\|\Delta m^{\mu\nu}\|_2^2 = c_{\mu\nu} D (1 + o(1)).$$

Then

$$\text{RMS}_0[\lambda(t) \langle x, \Delta m^{\mu\nu} \rangle] \sim \frac{e^{-t}}{\sqrt{\Delta_t}} \sqrt{c_{\mu\nu} D}. \quad (45)$$

and

$$t_{\text{rms}} = \frac{1}{2} \log \left(\frac{c_{\mu\nu} D}{T} \right) - \log L_{\text{rms}} + o(1). \quad (46)$$

Thus the familiar $\frac{1}{2} \log D$ scale is a consequence of accumulating weak evidence over D coordinates.

4.2 Planted recovery, the union bound, and the SNR

We now condition on the true latent component. Let μ^* be the pattern used to generate the observation and let

$$\hat{\mu}(x, t) \in \arg \max_{\mu} P(\mu \mid x, t).$$

Define the planted recovery margin

$$\kappa_{\star}(x, t) := \min_{\nu \neq \mu^*} \ell_{\mu^* \nu}(x, t).$$

Then $\hat{\mu} = \mu^*$ whenever $\kappa_{\star} > 0$. Moreover,

$$\mathbb{P}(\hat{\mu} \neq \mu^* \mid \mu^*) \leq \sum_{\nu \neq \mu^*} \mathbb{P}(\ell_{\mu^* \nu} \leq 0 \mid \mu^*). \quad (47)$$

This is the correct place to use the union bound: the planted index is fixed, and failure means that at least one competitor beats it. In particular, if every pairwise error is at most $\delta(t)$, then

$$\mathbb{P}(\hat{\mu} = \mu^* \mid \mu^*) \geq 1 - (M - 1)\delta(t). \quad (48)$$

An exact positivity identity. Before making a high-noise expansion, observe that the exact odds satisfy

$$\mathbb{E}_{x \sim P_t(\cdot \mid \mu^*)} [\ell_{\mu^* \nu}(x, t)] = \log \frac{P_0(\mu^*)}{P_0(\nu)} + D_{\text{KL}}(P_t(\cdot \mid \mu^*) \parallel P_t(\cdot \mid \nu)). \quad (49)$$

Therefore, under equal component priors, the exact expected log-odds are non-negative and are strictly positive whenever the two noised component laws differ. This conclusion does not require the vectors m^{μ^*} and m^{ν} to have equal norms.

First-order planted moments. Under the planted law,

$$x = e^{-t}a + \eta, \quad a \sim P_0(\cdot \mid \mu^*), \quad \eta \sim \mathcal{N}(0, \Delta_t I),$$

with a and η independent. Consequently,

$$\mathbb{E}[x_i \mid \mu^*] = e^{-t} m_i^{\mu^*}, \quad (50)$$

$$\text{Var}(x_i \mid \mu^*) = \Delta_t + e^{-2t} (1 - (m_i^{\mu^*})^2). \quad (51)$$

Applying these moments to the first-order approximation (37) gives

$$\mu_{\ell, \mu^* \nu}(t) := \mathbb{E}[\ell_{\mu^* \nu} \mid \mu^*] \approx \log \frac{P_0(\mu^*)}{P_0(\nu)} + \frac{e^{-2t}}{\Delta_t} \langle m^{\mu^*}, \Delta m^{\mu^* \nu} \rangle, \quad (52)$$

and

$$\sigma_{\ell, \mu^* \nu}^2(t) \approx \lambda(t)^2 \sum_{i=1}^D (\Delta m_i^{\mu^* \nu})^2 \left[\Delta_t + e^{-2t} (1 - (m_i^{\mu^*})^2) \right]. \quad (53)$$

The second term in (52) is not automatically positive for arbitrary unequal-norm templates. It is positive whenever $\langle m^{\mu^*}, m^{\nu} \rangle < \|m^{\mu^*}\|_2^2$. In the important equal-norm case,

$$\langle m^{\mu^*}, \Delta m^{\mu^* \nu} \rangle = \frac{1}{2} \|\Delta m^{\mu^* \nu}\|_2^2 > 0,$$

and this condition is automatic. Binary templates of common amplitude belong to this class. The exact KL identity (49) remains the definitive positivity statement outside the first-order approximation. If no single coordinate dominates the sum, a central-limit theorem gives

$$\ell_{\mu^* \nu} \mid \mu^* \approx \mathcal{N}(\mu_{\ell, \mu^* \nu}(t), \sigma_{\ell, \mu^* \nu}^2(t)).$$

Define the pairwise planted SNR

$$\text{SNR}_{\mu^* \nu}(t) := \frac{\mu_{\ell, \mu^* \nu}(t)}{\sigma_{\ell, \mu^* \nu}(t)}. \quad (54)$$

Under the Gaussian approximation,

$$\mathbb{P}(\ell_{\mu^*\nu} \leq 0 \mid \mu^*) \approx \Phi(-\text{SNR}_{\mu^*\nu}(t)).$$

Thus global recovery with probability at least $1 - \varepsilon$ is ensured approximately when every competitor satisfies

$$\text{SNR}_{\mu^*\nu}(t) \geq L_{\text{snr}}(M, \varepsilon) := \Phi^{-1}\left(1 - \frac{\varepsilon}{M-1}\right).$$

For small $\varepsilon/(M-1)$, $L_{\text{snr}} \sim \sqrt{2 \log((M-1)/\varepsilon)}$.

Closed form in the balanced high-noise regime. Assume equal priors and neglect the e^{-2t} correction in the variance bracket. Then

$$\text{SNR}_{\mu^*\nu}(t) \approx \frac{e^{-t}}{\sqrt{\Delta_t}} C_{\mu^*\nu}, \quad C_{\mu^*\nu} := \frac{\langle m^{\mu^*}, \Delta m^{\mu^*\nu} \rangle}{\|\Delta m^{\mu^*\nu}\|_2}. \quad (55)$$

The coefficient $C_{\mu^*\nu}$ is independent of t and T , but it generally *does depend on the dimension* through the template geometry. For equal-norm templates, $C_{\mu^*\nu} = \frac{1}{2} \|\Delta m^{\mu^*\nu}\|_2$. When $C_{\mu^*\nu} > 0$, solving the high-noise approximation $\text{SNR}_{\mu^*\nu} = L_{\text{snr}}$ gives

$$\boxed{t_{\text{rec}}^{\text{HN}}(L_{\text{snr}}; \mu^*, \nu) \approx \frac{1}{2} \log \left(1 + \frac{C_{\mu^*\nu}^2}{L_{\text{snr}}^2 T} \right)}. \quad (56)$$

This is a first-order, high-noise recovery formula, not an exact identity for arbitrary templates. If $C_{\mu^*\nu}^2 = \Theta(D)$, it scales as $\frac{1}{2} \log(D/T)$ up to an additive order-one term.

Remark 1 (RMS proxy versus planted recovery). *The RMS time (43) and the recovery time (56) share the same function $e^{-t}/\sqrt{\Delta_t}$, which explains their common logarithmic scaling. They are nevertheless different criteria and there is no general upper- or lower-bound relation between them unless their thresholds and geometric coefficients are matched. In the equal-norm case, $C_{\mu^*\nu}^2 = \|\Delta m^{\mu^*\nu}\|_2^2/4$, so using the same numerical threshold shifts the two times by an order-one constant (asymptotically, by $\log 2$).*

The Monte Carlo study should therefore report both posterior concentration and planted recovery rather than treating them as a single observable.

5 Binary templates: exact analysis

5.1 Exact pairwise log-odds in the binary setting

We work with $h_i^\mu = H\sigma_i^\mu$, $\sigma_i^\mu \in \{\pm 1\}$, $H > 0$, and the uniform prior $P_0(\mu) = 1/M$. In this section L denotes an SNR threshold; it should not be confused with the log-posterior margin threshold $c_{\varepsilon, M}$ in (36). The pairwise log-odds reduce to

$$\ell_{\mu\nu}(x, t) = S_\mu(x, t) - S_\nu(x, t),$$

with

$$S_\mu(x, t) = \sum_{i=1}^D \left[\log \cosh(\beta H \sigma_i^\mu + \lambda(t) x_i) - \log \cosh(\beta H) \right].$$

Step 1: only disagreeing sites contribute. At each site i , the pair $(\sigma_i^\mu, \sigma_i^\nu)$ falls into one of two cases. *Case A:* $\sigma_i^\mu = \sigma_i^\nu$. The terms in S_μ and S_ν at site i are identical, so their difference is exactly zero. *Case B:* $\sigma_i^\mu \neq \sigma_i^\nu$. Since both σ_i^μ and σ_i^ν are in $\{\pm 1\}$, disagreement means $\sigma_i^\nu = -\sigma_i^\mu$. The two arguments become $\beta H \sigma_i^\mu + \lambda x_i$ and $-\beta H \sigma_i^\mu + \lambda x_i$. Let $\mathcal{D}_{\mu\nu} := \{i : \sigma_i^\mu \neq \sigma_i^\nu\}$ denote the set of disagreeing sites. Then

$$\ell_{\mu\nu}(x, t) = \sum_{i \in \mathcal{D}_{\mu\nu}} \left[\log \cosh(\beta H \sigma_i^\mu + \lambda x_i) - \log \cosh(-\beta H \sigma_i^\mu + \lambda x_i) \right]. \quad (57)$$

Step 2: simplify each per-site term. Fix a disagreeing site i and define

$$a_i := \beta H \sigma_i^\mu, \quad b_i := \lambda(t) x_i.$$

We need $\log \cosh(a_i + b_i) - \log \cosh(-a_i + b_i)$. Since $\cosh$ is an even function, $\cosh(-a_i + b_i) = \cosh(a_i - b_i)$, so this equals

$$\log \cosh(a_i + b_i) - \log \cosh(a_i - b_i) = \log \frac{\cosh(a_i + b_i)}{\cosh(a_i - b_i)}.$$

Using hyperbolic functions' identities:

$$\cosh(A + B) = \cosh A \cosh B + \sinh A \sinh B, \quad \cosh(A - B) = \cosh A \cosh B - \sinh A \sinh B.$$

Dividing numerator and denominator by $\cosh a_i \cosh b_i$:

$$\frac{\cosh(a_i + b_i)}{\cosh(a_i - b_i)} = \frac{1 + \tanh a_i \tanh b_i}{1 - \tanh a_i \tanh b_i}.$$

Taking the logarithm and using the identity $\log \frac{1+z}{1-z} = 2 \operatorname{artanh}(z)$ for $|z| < 1$ yields

$$\log \cosh(a_i + b_i) - \log \cosh(a_i - b_i) = 2 \operatorname{artanh}(\tanh a_i \cdot \tanh b_i).$$

Substituting back $a_i = \beta H \sigma_i^\mu$ and $b_i = \lambda(t) x_i$, and using $\tanh(\beta H \sigma_i^\mu) = \sigma_i^\mu \tanh(\beta H) = \sigma_i^\mu m_0$ (since $\tanh$ is odd), each disagreeing site contributes

$$2 \operatorname{artanh}(\sigma_i^\mu m_0 \cdot \tanh(\lambda(t) x_i)).$$

Summing over all disagreeing sites gives the exact, closed-form pairwise log-odds:

$$\boxed{\ell_{\mu\nu}(x, t) = 2 \sum_{i \in \mathcal{D}_{\mu\nu}} \operatorname{artanh}(\sigma_i^\mu m_0 \cdot \tanh(\lambda(t) x_i))}. \quad (58)$$

Recovering the Taylor approximation as a check. When $\lambda(t)$ is small, $\tanh(\lambda x_i) \approx \lambda x_i$ and $\operatorname{artanh}(z) \approx z$ for small z . Hence each term becomes approximately $2\sigma_i^\mu m_0 \lambda x_i$, and summing over disagreeing sites:

$$\ell_{\mu\nu}(x, t) \approx 2\lambda(t) m_0 \sum_{i \in \mathcal{D}_{\mu\nu}} \sigma_i^\mu x_i.$$

Since $\Delta m_i^{\mu\nu} = m_0(\sigma_i^\mu - \sigma_i^\nu) = 2m_0\sigma_i^\mu$ at disagreeing sites and zero elsewhere, $2m_0 \sum_{i \in \mathcal{D}} \sigma_i^\mu x_i = \sum_{i=1}^D \Delta m_i^{\mu\nu} x_i = \langle x, \Delta m^{\mu\nu} \rangle$, recovering exactly the leading-order formula $\ell_{\mu\nu} \approx \lambda(t) \langle x, \Delta m^{\mu\nu} \rangle$.

5.2 Exact SNR under the planted measure

We now study the exact log-odds (58) under the planted measure, without any Taylor expansion.

Per-site independence. Conditional on the planted pattern $\mu^\star$, the x_i are independent (since the a_i are independent under $P_0(\cdot \mid \mu^\star)$ and the noise η_i are independent Gaussians). Therefore the per-site contributions

$$g_i := 2 \operatorname{artanh}(\sigma_i^{\mu^\star} m_0 \cdot \tanh(\lambda(t) x_i)), \quad i \in \mathcal{D}_{\mu^\star \nu},$$

are mutually independent random variables. Define

$$\mu_g(t) := \mathbb{E}[g_i \mid \mu^\star], \quad \sigma_g^2(t) := \operatorname{Var}(g_i \mid \mu^\star). \quad (59)$$

Since all disagreeing sites have the same distributional structure (the distribution of x_i depends on $\sigma_i^{\mu^\star}$ only through $m_i^{\mu^\star} = \sigma_i^{\mu^\star} m_0$, and the per-site contribution involves $\sigma_i^{\mu^\star}$ in a way that makes g_i identically distributed across disagreeing sites—see below), these moments are site-independent.

Why all disagreeing sites are identically distributed

At a disagreeing site, $\sigma_i^\nu = -\sigma_i^{\mu^*}$. Write $\epsilon_i := \sigma_i^{\mu^*} \in \{\pm 1\}$. Then $g_i = 2 \operatorname{artanh}(\epsilon_i m_0 \cdot \tanh(\lambda x_i))$. Under the planted measure, $x_i = e^{-t} a_i + \eta_i$ with $a_i \sim P_0(\cdot \mid \mu^*)$, so $\mathbb{E}[a_i \mid \mu^*] = \epsilon_i m_0$. Define $a'_i := \epsilon_i a_i$ and $x'_i := \epsilon_i x_i$. Then $a'_i \in \{\pm 1\}$ with $\mathbb{E}[a'_i] = m_0$ (independent of i), $x'_i = e^{-t} a'_i + \epsilon_i \eta_i$, and since η_i is symmetric, $\epsilon_i \eta_i \sim \mathcal{N}(0, \Delta_t)$. So x'_i has the same distribution for every disagreeing site. Moreover,

$$g_i = 2 \operatorname{artanh}(m_0 \cdot \tanh(\lambda x'_i)),$$

which is the same function of x'_i for every i . Hence the g_i are identically distributed.

Exact mean and variance of $\ell_{\mu^* \nu}$. Since the $d_{\mu^* \nu}$ terms are i.i.d.:

$$\mathbb{E}[\ell_{\mu^* \nu} \mid \mu^*] = d_{\mu^* \nu} \mu_g(t), \quad \operatorname{Var}(\ell_{\mu^* \nu} \mid \mu^*) = d_{\mu^* \nu} \sigma_g^2(t), \quad (60)$$

and the exact SNR is

$$\operatorname{SNR}_{\text{exact}}(t) = \frac{d_{\mu^* \nu} \mu_g(t)}{\sqrt{d_{\mu^* \nu}} \sigma_g(t)} = \sqrt{d_{\mu^* \nu}} \frac{\mu_g(t)}{\sigma_g(t)}. \quad (61)$$

The per-site moments $\mu_g(t)$ and $\sigma_g(t)$ are functions of βH , T , and t only, given by

$$\mu_g(t) = \mathbb{E}\left[2 \operatorname{artanh}(m_0 \cdot \tanh(\lambda(t) x'))\right], \quad (62)$$

$$\sigma_g^2(t) = \mathbb{E}\left[\left(2 \operatorname{artanh}(m_0 \cdot \tanh(\lambda(t) x'))\right)^2\right] - \mu_g(t)^2, \quad (63)$$

where $x' = e^{-t} a' + \eta$, $a' \in \{\pm 1\}$ with $\mathbb{P}(a' = +1) = \frac{1+m_0}{2}$, $\eta \sim \mathcal{N}(0, \Delta_t)$ independent of a' . These expectations do not simplify to elementary closed forms, but they are one-dimensional integrals (over the Gaussian η , with a discrete sum over $a' \in \{\pm 1\}$) and can be evaluated numerically to arbitrary precision for any βH , T , and t .

Moment-based and error-based selection times. The per-site moments above are exact, so the moment-based SNR crossover is defined implicitly by

$$\sqrt{d_{\mu^* \nu}} \frac{\mu_g(t)}{\sigma_g(t)} = L. \quad (64)$$

It can be solved numerically without a Taylor expansion. Its interpretation as a Gaussian error threshold, however, requires $d_{\mu^* \nu}$ to be large enough for a CLT. For arbitrary d , the genuinely exact pairwise Bayes error (with a random tie break) is

$$p_{\text{err}}(t; d) := \mathbb{P}\left(\sum_{r=1}^d g_r(t) < 0 \mid \mu^*\right) + \frac{1}{2} \mathbb{P}\left(\sum_{r=1}^d g_r(t) = 0 \mid \mu^*\right). \quad (65)$$

For $t > 0$ the tie event has probability zero; it matters at the discrete endpoint $t = 0$. The probability can be evaluated by one-dimensional sampling followed by convolution, or directly by Monte Carlo. Since the OU semigroup only adds information-destroying noise as forward time grows, the optimal pairwise Bayes error is non-decreasing in t (a data-processing statement). Hence an error-based crossover can be defined by

$$t_{\text{err}}(\varepsilon; d) := \sup\{t \geq 0 : p_{\text{err}}(t; d) \leq \varepsilon\},$$

whenever the set is non-empty. There can be an irreducible error when d is fixed. At $t = 0$ the clean configuration is observed and

$$\ell_{\mu^* \nu}(a, 0) = 2\beta H \sum_{r=1}^d B_r, \quad \mathbb{P}(B_r = 1) = \frac{1+m_0}{2},$$

so $p_{\text{err}}(0; d)$ is a binomial tail rather than zero. The endpoint moment ratio is

$$\operatorname{SNR}_{\text{exact}}(0) = \sqrt{d} \frac{m_0}{\sqrt{1-m_0^2}}.$$

Therefore a chosen moment-SNR threshold should first be checked against this endpoint value. For small d , the exact error criterion (65) is more reliable than treating the SNR as a Gaussian error probability.

5.3 First-order and high-noise asymptotics

The exact one-site moments $\mu_g(t)$ and $\sigma_g(t)$ generally require numerical integration. A useful analytic benchmark is obtained by linearizing the exact contribution, $g \approx 2m_0\lambda(t)x'$. Retaining the full variance of x' gives

$$\text{SNR}^{(1)}(t; d) = \frac{m_0\sqrt{d}e^{-t}}{\sqrt{\Delta_t + e^{-2t}(1 - m_0^2)}}. \quad (66)$$

Solving $\text{SNR}^{(1)}(t; d) = L$ yields

$$t_{\text{rec}}^{(1)}(L; d) = \frac{1}{2} \log \left(1 + \frac{m_0^2 d - L^2(1 - m_0^2)}{L^2 T} \right), \quad (67)$$

provided $m_0^2 d > L^2(1 - m_0^2)$. This feasibility condition is exactly $\text{SNR}_{\text{exact}}(0) > L$ at the level of the linearized moment criterion. In the genuinely high-noise regime the spin-variance correction $e^{-2t}(1 - m_0^2)$ is negligible compared with Δ_t . Then

$$\mu_g(t) \approx \frac{2m_0^2 e^{-2t}}{\Delta_t}, \quad \sigma_g^2(t) \approx \frac{4m_0^2 e^{-2t}}{\Delta_t}, \quad \frac{\mu_g(t)}{\sigma_g(t)} \approx \frac{m_0 e^{-t}}{\sqrt{\Delta_t}},$$

and the simpler crossover is

$$t_{\text{rec}}^{\text{HN}}(L; d) \approx \frac{1}{2} \log \left(1 + \frac{m_0^2 d}{L^2 T} \right), \quad m_0 = \tanh(\beta H). \quad (68)$$

Equation (68) is a high-noise asymptotic, whereas (67) retains the leading endpoint correction. In particular, the formal positive value returned by (68) does not guarantee that the requested finite- d accuracy is attainable: the endpoint error (65), or at least the feasibility condition below (67), must also be checked. Both analytic approximations should be compared with the exact moment equation (64) and the exact error curve (65) in the simulations.

From pairwise to global selection

This is still a pairwise statement: different competitors ν yield different selection times, with closer patterns (smaller $d_{\mu^* \nu}$) requiring more denoising. For μ^* to dominate *all* competitors simultaneously, the binding constraint comes from the nearest rival. By the same union-bound logic as in Section 4, global selection at confidence $1 - \varepsilon$ requires the pairwise SNR to exceed $L = \Phi^{-1} \left(1 - \frac{\varepsilon}{M-1} \right)$ for the hardest pair, giving the bottleneck timescale

$$t_{\text{rec}}^{\text{bottleneck}} \approx \frac{1}{2} \log \left(1 + \frac{m_0^2 d_{\min}}{L^2 T} \right), \quad d_{\min} := \min_{\nu \neq \mu^*} d_{\mu^* \nu}. \quad (69)$$

For generic random binary patterns, each site disagrees independently with probability $1/2$, so $d \approx D/2$. Substituting:

$$t_{\text{rec}}^{\text{HN}} \approx \frac{1}{2} \log \left(1 + \frac{\tanh^2(\beta H) D}{2L^2 T} \right) \quad (\text{generic binary patterns}). \quad (70)$$

If $\tanh^2(\beta H) D / (2L^2 T) \gg 1$, the $1+$ may be dropped at leading order. This is the binary-template analogue of the general $\frac{1}{2} \log(D/T)$ scaling. Two explicit physical parameters now appear: $\tanh^2(\beta H)$ quantifies how strongly the inverse temperature polarises the spins (maximal at $\beta H \rightarrow \infty$, vanishing at $\beta H \rightarrow 0$), and d replaces D for non-generic patterns. The bottleneck pair—the one that determines the overall selection time—is the pair with the smallest Hamming distance:

$$d_{\min} := \min_{\mu \neq \nu} d_{\mu \nu}. \quad (71)$$

When $d_{\min}$ is small (close patterns), this can be much smaller than $\frac{1}{2} \log(D/T)$, meaning the reverse diffusion must denoise much further before distinguishing the closest pair.

5.4 Approximations in the general setting and their impact on correctness

The selection timescale derived is a product of various approximations which could impact any analysis that claims to be general:

The Taylor expansion of $S_\mu(x, t)$. The exact scalar log-evidence

$$S_\mu(x, t) = \sum_i \left[\log \cosh(\beta h_i^\mu + \lambda(t)x_i) - \log \cosh(\beta h_i^\mu) \right]$$

was replaced by its first-order term

$$S_\mu \approx \lambda(t) \langle x, m^\mu \rangle.$$

A sufficient coordinatewise condition is

$$\lambda(t) \sqrt{\Delta_t \log D} \ll 1.$$

In the binary model, evaluating this quantity at the high-noise recovery scale makes it proportional to $L\sqrt{\log D}/(m_0\sqrt{d})$. Thus the expansion is controlled for $d \gg \log D$, but not automatically for very close patterns. For general templates, the second-order pairwise correction

$$\frac{\lambda(t)^2}{2} \sum_i x_i^2 ((m_i^\nu)^2 - (m_i^\mu)^2)$$

need not cancel and may shift the odds.

The marginal approximation $x \approx \mathcal{N}(0, \Delta_t I)$. This replaces the planted distribution $x_i = e^{-t}a_i + \eta_i$ (which has mean $e^{-t}m_i^{\mu^*}$ per coordinate) by a centred Gaussian. As discussed in Section 4.2, the per-coordinate error is $O(e^{-t})$, negligible for each site, but coherently accumulated over D sites it produces the $O(D)$ mean of $\ell_{\mu^*\nu}$ that drives selection. The marginal approximation gives the leading high-noise variance but misses the planted mean, and it is the planted-measure analysis that restores the directional signal.

Gaussianity of $\ell_{\mu^*\nu}$ via the CLT. The pairwise log-odds are a sum of D independent per-site terms. The CLT applies provided no single site dominates, i.e. $\|\Delta m^{\mu\nu}\|_\infty / \|\Delta m^{\mu\nu}\|_2 \rightarrow 0$. This fails when two patterns differ at only $O(1)$ sites: the sum has too few terms for the Gaussian approximation, and the tail bounds that control error probabilities become unreliable.

The variance upper bound. The exact conditional variance involves a site-dependent bracket $\Delta_t + e^{-2t}(1 - (m_i^{\mu^*})^2)$, which was replaced by its maximum $\Delta_t + e^{-2t}$. For general templates with heterogeneous field strengths this bound can be quite loose, inflating the variance and underestimating the SNR.

What binary templates fix. Specialising to $h_i^\mu = H\sigma_i^\mu$ with $\sigma_i^\mu \in \{\pm 1\}$ and $H > 0$ eliminates or tightens most of these issues:

- The static prior becomes exactly uniform, $P_0(\mu) = 1/M$, so the log-prior ratio $\log \frac{P_0(\mu)}{P_0(\nu)}$ vanishes identically. There is no thermodynamic dominance; we are squarely in the balanced regime.
- The magnetization takes the form $m_i^\mu = m_0\sigma_i^\mu$ with $m_0 := \tanh(\beta H)$ a universal scalar. In particular, $(m_i^\mu)^2 = m_0^2$ is the same at every site and for every pattern, so the second-order Taylor correction in $S_\mu - S_\nu$ cancels exactly. The quadratic term in the pairwise difference vanishes, so the first neglected term is cubic in λx .
- The variance bound becomes tight: the site-dependent bracket reduces to $\Delta_t + e^{-2t}(1 - m_0^2)$, which is the same for every site.
- The geometry of the pattern space reduces to Hamming combinatorics: $\|\Delta m^{\mu\nu}\|_2^2 = 4m_0^2 d_{\mu\nu}$, $\langle m^{\mu^*}, \Delta m^{\mu^*\nu} \rangle = 2m_0^2 d_{\mu^*\nu}$.

What remains. Two approximations survive even in the binary setting. First, the Taylor expansion itself: although the second-order term vanishes in $S_\mu - S_\nu$, the third-order and higher terms do not. For moderate D or small Hamming distance d , these can be significant at t_{sel} . Second, the Gaussianity assumption via the CLT requires $d_{\mu\nu} \gg 1$; for patterns differing at $O(1)$ sites it fails. Both issues are addressed by the exact pairwise log-odds derived above; for small d , the exact error probability (65) should replace the Gaussian tail approximation.

6 The exact diffusion score, denoiser, and curvature

We now move from posterior odds to the vector-valued diffusion score. This supplies an exact denoising identity and a complementary local curvature criterion for speciation.

6.1 Deriving the score

The marginal density of x at time t is

$$P_t(x) = \sum_{\mu=1}^M P_0(\mu) P_t(x \mid \mu).$$

Taking the gradient with respect to x :

$$\nabla_x \log P_t(x) = \frac{\nabla_x P_t(x)}{P_t(x)} = \frac{\sum_{\mu=1}^M P_0(\mu) \nabla_x P_t(x \mid \mu)}{\sum_{\mu=1}^M P_0(\mu) P_t(x \mid \mu)}.$$

recognising the posterior weights $P(\mu \mid x, t) = P_0(\mu) P_t(x \mid \mu) / P_t(x)$ reads:

$$\nabla_x \log P_t(x) = \sum_{\mu=1}^M P(\mu \mid x, t) \nabla_x \log P_t(x \mid \mu). \quad (72)$$

highlighting how *the score is the posterior-weighted average of the per-pattern scores*.

6.2 Per-pattern score

From the exact log-likelihood (26),

$$\log P_t(x \mid \mu) = \underbrace{-\frac{D}{2} \log(2\pi\Delta_t) - \frac{\|x\|^2}{2\Delta_t} - \frac{De^{-2t}}{2\Delta_t}}_{\log c(x,t)} + \sum_{i=1}^D \left[\log \cosh(\beta h_i^\mu + \lambda(t)x_i) - \log \cosh(\beta h_i^\mu) \right].$$

Differentiating with respect to x_j :

$$\frac{\partial}{\partial x_j} \log P_t(x \mid \mu) = -\frac{x_j}{\Delta_t} + \lambda(t) \tanh(\beta h_j^\mu + \lambda(t)x_j).$$

The first term comes from $-\|x\|^2/(2\Delta_t)$ in $\log c(x, t)$; it is independent of μ . The second term is the μ -dependent contribution, and it admits a direct probabilistic interpretation.

The per-pattern denoiser. Define, at each coordinate j ,

$$\hat{a}_j^\mu(x, t) := \tanh(\beta h_j^\mu + \lambda(t)x_j). \quad (73)$$

This is the *posterior mean of a_j given μ and x* .

Why so?

Recall that conditional on μ , the spin a_j has prior $P_0(a_j | \mu) \propto e^{\beta h_j^\mu a_j}$ (a ± 1 -valued random variable with bias βh_j^μ), and the observation x_j provides a Gaussian likelihood $p(x_j | a_j, t) \propto \exp(a_j \lambda(t) x_j)$ (up to terms independent of a_j ; this follows from expanding $-(x_j - e^{-t} a_j)^2 / (2\Delta_t)$). Combining prior and likelihood by Bayes:

$$P(a_j = +1 | \mu, x_j, t) = \frac{e^{\beta h_j^\mu + \lambda(t) x_j}}{e^{\beta h_j^\mu + \lambda(t) x_j} + e^{-\beta h_j^\mu - \lambda(t) x_j}} = \frac{1}{1 + e^{-2(\beta h_j^\mu + \lambda(t) x_j)}},$$

and the posterior mean is

$$\mathbb{E}[a_j | \mu, x, t] = P(a_j = +1 | \mu, x_j, t) - P(a_j = -1 | \mu, x_j, t) = \tanh(\beta h_j^\mu + \lambda(t) x_j) = \hat{a}_j^\mu(x, t).$$

So $\hat{a}^\mu(x, t)$ is the best guess of the clean configuration a if one knew the pattern were μ —a *pattern-specific denoiser*.

The per-pattern score is therefore

$$\nabla_x \log P_t(x | \mu) = -\frac{x}{\Delta_t} + \lambda(t) \hat{a}^\mu(x, t). \quad (74)$$

6.3 The full score and the posterior-averaged denoiser

Plugging (74) into (72):

$$\begin{aligned} \nabla_x \log P_t(x) &= \sum_{\mu=1}^M P(\mu | x, t) \left[-\frac{x}{\Delta_t} + \lambda(t) \hat{a}^\mu(x, t) \right] \\ &= -\frac{x}{\Delta_t} + \lambda(t) \underbrace{\sum_{\mu=1}^M P(\mu | x, t) \hat{a}^\mu(x, t)}_{=: \hat{a}(x, t)}, \end{aligned} \quad (75)$$

where the $-x/\Delta_t$ term factors out because it is μ -independent, and we define

$$\hat{a}(x, t) := \sum_{\mu=1}^M P(\mu | x, t) \hat{a}^\mu(x, t) \quad (76)$$

as the **posterior-averaged denoiser**: the best estimate of the clean configuration a .

The score as a denoising instruction

The exact score takes the form

$$\boxed{\nabla_x \log P_t(x) = -\frac{x}{\Delta_t} + \lambda(t) \hat{a}(x, t).} \quad (77)$$

Equivalently,

$$\nabla_x \log P_t(x) = -\frac{x - e^{-t} \hat{a}(x, t)}{\Delta_t}.$$

This is the exact denoising identity: the score compares the current state x with the posterior prediction $e^{-t} \hat{a}(x, t)$ of its noised clean ancestor. The score is one contribution to the reverse SDE drift; the OU drift and the chosen time parametrization must also be included when writing the full reverse dynamics.

6.4 How selection appears in the score

The decomposition (75) makes pattern selection visible in the score itself. It is useful to write the exact identity as

$$\nabla_x \log P_t(x) = -\frac{x - e^{-t}\hat{a}(x, t)}{\Delta_t},$$

where $\hat{a}(x, t) = \mathbb{E}[a \mid x, t]$. Thus the score points from x toward the posterior prediction $e^{-t}\hat{a}(x, t)$ of the noised clean signal. The score enters the reverse SDE (or probability flow ODE) with a convention-dependent coefficient; it should not by itself be identified with the full reverse drift. At high noise, $P(\mu \mid x, t) \approx P_0(\mu)$ and $\hat{a}(x, t)$ averages several pattern-specific denoisers. After posterior selection, $P(\mu^* \mid x, t) \approx 1$ and $\hat{a}(x, t) \approx \hat{a}^{\mu^*}(x, t)$. The log-odds analysis describes this global posterior concentration. The next subsection studies a related but different local phenomenon: the loss of concavity of $\log P_t$ at the balanced high-noise point.

6.5 Curvature instability of the score: the Curie–Weiss spinodal

Let

$$s_t(x) := \nabla_x \log P_t(x), \quad J(t) := \nabla_x^2 \log P_t(0).$$

Assume the component family is balanced so that

$$\sum_{\mu=1}^M P_0(\mu) m^\mu = 0. \quad (78)$$

Since $S_\mu(0, t) = 0$, one has $P(\mu \mid 0, t) = P_0(\mu)$ and therefore $\hat{a}(0, t) = 0$. Hence $s_t(0) = 0$. The relevant notion of stability is local curvature. For a small perturbation $x = \varepsilon v$,

$$s_t(\varepsilon v) = \varepsilon J(t)v + O(\varepsilon^2).$$

If every eigenvalue of $J(t)$ is negative, the origin is a strict local maximum of P_t (equivalently, a local minimum of $-\log P_t$), and the score points back toward the origin in every direction. When the largest eigenvalue of $J(t)$ reaches zero, one quadratic restoring direction disappears. After it becomes positive, the origin is no longer a local maximum along that eigendirection. This curvature crossing is the Landau-type spinodal/speciation criterion; it is not, by itself, the same event as posterior recovery of a particular μ^* . At $x = 0$,

$$P(\mu \mid 0, t) = P_0(\mu), \quad \hat{a}_j^\mu(0, t) = m_j^\mu := \tanh(\beta h_j^\mu).$$

We now compute $J(t)$ explicitly.

Jacobian at $x = 0$ (intermediate calculations). From (77),

$$\nabla_x \log P_t(x) = -\frac{x}{\Delta_t} + \lambda(t) \hat{a}(x, t), \quad \hat{a}_j(x, t) = \sum_{\mu=1}^M P(\mu \mid x, t) \hat{a}_j^\mu(x, t).$$

Differentiate componentwise:

$$J_{jk}(t) := \frac{\partial}{\partial x_k} (\partial_{x_j} \log P_t(x)) \Big|_{x=0} = -\frac{\delta_{jk}}{\Delta_t} + \lambda(t) \frac{\partial \hat{a}_j}{\partial x_k}(0, t). \quad (79)$$

So we must compute $\partial_{x_k} \hat{a}_j(0, t)$. **Step 1: derivative of the posterior weights.** Since

$$P(\mu \mid x, t) = \frac{P_0(\mu) e^{S_\mu(x, t)}}{\sum_\nu P_0(\nu) e^{S_\nu(x, t)}},$$

we can differentiate $\log P(\mu \mid x, t)$:

$$\partial_{x_k} \log P(\mu \mid x, t) = \partial_{x_k} S_\mu(x, t) - \sum_\nu P(\nu \mid x, t) \partial_{x_k} S_\nu(x, t).$$

Exponentiating back,

$$\partial_{x_k} P(\mu | x, t) = P(\mu | x, t) \left(\partial_{x_k} S_\mu(x, t) - \sum_{\nu} P(\nu | x, t) \partial_{x_k} S_\nu(x, t) \right). \quad (80)$$

From the definition of S_μ ,

$$S_\mu(x, t) = \sum_{i=1}^D \left[\log \cosh(\beta h_i^\mu + \lambda x_i) - \log \cosh(\beta h_i^\mu) \right],$$

and therefore only the $i = k$ term depends on x_k , giving

$$\frac{\partial S_\mu}{\partial x_k}(x, t) = \lambda(t) \tanh(\beta h_k^\mu + \lambda(t)x_k) = \lambda(t) \hat{a}_k^\mu(x, t). \quad (81)$$

Evaluating at $x = 0$ yields $\partial_{x_k} S_\mu(0, t) = \lambda(t) m_k^\mu$ and $P(\mu | 0, t) = P_0(\mu)$, hence

$$\partial_{x_k} P(\mu | 0, t) = P_0(\mu) \left(\lambda(t) m_k^\mu - \lambda(t) \sum_{\nu} P_0(\nu) m_k^\nu \right) = \lambda(t) P_0(\mu) \left(m_k^\mu - \mathbb{E}_{P_0}[m_k^\nu] \right). \quad (82)$$

Step 2: derivative of the per-pattern denoiser. Recall $\hat{a}_j^\mu(x, t) = \tanh(\beta h_j^\mu + \lambda(t)x_j)$. Thus

$$\frac{\partial \hat{a}_j^\mu}{\partial x_k}(x, t) = \delta_{jk} \lambda(t) \operatorname{sech}^2(\beta h_j^\mu + \lambda(t)x_j) = \delta_{jk} \lambda(t) \left(1 - (\hat{a}_j^\mu(x, t))^2 \right). \quad (83)$$

At $x = 0$, $\hat{a}_j^\mu(0, t) = m_j^\mu$, so

$$\frac{\partial \hat{a}_j^\mu}{\partial x_k}(0, t) = \delta_{jk} \lambda(t) (1 - (m_j^\mu)^2). \quad (84)$$

Step 3: combine by the product rule. Differentiate $\hat{a}_j(x, t) = \sum_{\mu} P(\mu | x, t) \hat{a}_j^\mu(x, t)$:

$$\begin{aligned} \frac{\partial \hat{a}_j}{\partial x_k}(0, t) &= \sum_{\mu} \left(\partial_{x_k} P(\mu | 0, t) \right) \hat{a}_j^\mu(0, t) + \sum_{\mu} P(\mu | 0, t) \frac{\partial \hat{a}_j^\mu}{\partial x_k}(0, t) \\ &= \sum_{\mu} \left[\lambda(t) P_0(\mu) (m_k^\mu - \mathbb{E}_{P_0}[m_k]) \right] m_j^\mu + \sum_{\mu} P_0(\mu) \delta_{jk} \lambda(t) (1 - (m_j^\mu)^2) \\ &= \lambda(t) \left(\mathbb{E}_{P_0}[m_j^\mu m_k^\mu] - \mathbb{E}_{P_0}[m_j^\mu] \mathbb{E}_{P_0}[m_k^\mu] \right) + \delta_{jk} \lambda(t) \mathbb{E}_{P_0}[1 - (m_j^\mu)^2] \\ &= \lambda(t) \operatorname{Cov}_{P_0}(m_j^\mu, m_k^\mu) + \delta_{jk} \lambda(t) (1 - \overline{m_j^2}), \end{aligned} \quad (85)$$

where we introduced the notation

$$\overline{m_j^2} := \mathbb{E}_{P_0}[(m_j^\mu)^2]. \quad (86)$$

Step 4: plug into the Jacobian. Substituting (85) into (79) yields

$$\begin{aligned} J_{jk}(t) &= -\frac{\delta_{jk}}{\Delta_t} + \lambda(t) \left[\lambda(t) \operatorname{Cov}_{P_0}(m_j^\mu, m_k^\mu) + \delta_{jk} \lambda(t) (1 - \overline{m_j^2}) \right] \\ &= -\frac{\delta_{jk}}{\Delta_t} + \lambda(t)^2 \left[\delta_{jk} (1 - \overline{m_j^2}) + \operatorname{Cov}_{P_0}(m_j^\mu, m_k^\mu) \right] \\ &= \begin{cases} -\frac{1}{\Delta_t} + \lambda(t)^2 \left[(1 - \overline{m_j^2}) + \operatorname{Var}_{P_0}(m_j^\mu) \right], & j = k, \\ \lambda(t)^2 \operatorname{Cov}_{P_0}(m_j^\mu, m_k^\mu), & j \neq k, \end{cases} \end{aligned} \quad (87)$$

where we used $\operatorname{Cov}(X, X) = \operatorname{Var}(X)$. The bracket in (87) has a useful probabilistic interpretation. By the law of total covariance,

$$\Sigma_a := \operatorname{Cov}_{P_0}(a) = \operatorname{diag}(1 - \overline{m_1^2}, \dots, 1 - \overline{m_D^2}) + \operatorname{Cov}_{P_0}(m^\mu). \quad (88)$$

Consequently, under the balance condition (78),

$$J(t) = -\frac{1}{\Delta_t}I + \frac{e^{-2t}}{\Delta_t^2}\Sigma_a. \quad (89)$$

This is exactly the covariance form expected from a large-time Landau expansion: the isotropic Gaussian term supplies negative curvature, whereas directions with large data variance receive the strongest positive correction.

Binary templates: operator form. Take

$$h_i^\mu = H\sigma_i^\mu, \quad m_0 := \tanh(\beta H), \quad P_0(\mu) = \frac{1}{M},$$

and assume sitewise balance, $\sum_{\mu=1}^M \sigma_i^\mu = 0$ for every i . Then $m_i^\mu = m_0\sigma_i^\mu$ and

$$\Sigma_a = (1 - m_0^2)I + m_0^2K, \quad K := \frac{1}{M} \sum_{\mu=1}^M \sigma^\mu \sigma^{\mu\top}.$$

Therefore

$$J(t) = \left(-\frac{1}{\Delta_t} + \lambda(t)^2(1 - m_0^2) \right) I + \lambda(t)^2 m_0^2 K. \quad (90)$$

Let $\mathcal{S}_{\text{pat}} := \text{span}\{\sigma^1, \dots, \sigma^M\}$. For every $v \in \mathbb{R}^D$,

$$Kv = \frac{1}{M} \sum_{\mu=1}^M \sigma^\mu \langle \sigma^\mu, v \rangle,$$

so $\text{Im}(K) \subseteq \mathcal{S}_{\text{pat}}$. Moreover,

$$v^\top Kv = \frac{1}{M} \sum_{\mu=1}^M \langle v, \sigma^\mu \rangle^2,$$

which shows that $\ker(K) = \mathcal{S}_{\text{pat}}^\perp$. Since K is symmetric,

$$\mathbb{R}^D = \ker(K) \oplus \text{Im}(K) = \mathcal{S}_{\text{pat}}^\perp \oplus \mathcal{S}_{\text{pat}}. \quad (91)$$

Thus the nonzero-eigenvalue eigenspace of K is exactly the subspace spanned by the pre-existing templates. Because $J(t) = a(t)I + b(t)K$, the two matrices have the same eigenvectors. Indeed, if $Kv = \gamma v$, then

$$J(t)v = (a(t) + b(t)\gamma)v.$$

Let

$$\gamma_1 \geq \dots \geq \gamma_{\text{rank}(K)} > 0$$

be the nonzero eigenvalues of K . The curvature eigenvalues are

$$\Lambda_{\text{bulk}}(t) = -\frac{1}{\Delta_t} + \lambda(t)^2(1 - m_0^2)$$

on $\ker(K)$, and

$$\Lambda_r(t) = -\frac{1}{\Delta_t} + \lambda(t)^2(1 - m_0^2 + m_0^2\gamma_r)$$

on the corresponding template-aligned eigendirection. Hence

$$\text{spec } J(t) = \underbrace{\{\Lambda_{\text{bulk}}(t)\}}_{D - \text{rank}(K) \text{ copies}} \cup \underbrace{\{\Lambda_r(t) : 1 \leq r \leq \text{rank}(K)\}}_{\text{template subspace}}. \quad (92)$$

The bulk modes may also lose concavity at sufficiently small t ; the point is that any template mode with $\gamma_r > 0$ crosses earlier than the bulk because it has the additional positive term $m_0^2\gamma_r$. The first class-level instability is therefore controlled by the largest covariance eigenvalue.

Exact mode-dependent spinodal times. Define

$$\zeta_r := 1 - m_0^2 + m_0^2 \gamma_r, \quad \zeta_{\text{bulk}} := 1 - m_0^2.$$

Since $\lambda(t)^2 = e^{-2t}/\Delta_t^2$, the condition $\Lambda_r(t) = 0$ is

$$e^{-2t} \zeta_r = \Delta_t = T(1 - e^{-2t}).$$

Writing $u = e^{-2t}$ gives $u\zeta_r = T(1 - u)$, and therefore

$$t_{\text{spin},r} = \frac{1}{2} \log \left(1 + \frac{\zeta_r}{T} \right). \quad (93)$$

The first curvature instability is obtained from γ_1 :

$$t_{\text{spin}} = \frac{1}{2} \log \left(1 + \frac{1 + m_0^2(\gamma_1 - 1)}{T} \right). \quad (94)$$

No approximation $\Delta_t \approx T$ is used here. If $\zeta_1 \gg T$, then $t_{\text{spin}} = \frac{1}{2} \log(\zeta_1/T) + o(1)$. In the normalization $T = 1$ and for large ζ_1 , this is equivalent to the familiar speciation criterion $\zeta_1 e^{-2t_{\text{spin}}} \asymp 1$.

Preferred direction and degeneracy. For a perturbation along an eigenvector v_r , $s_t(\varepsilon v_r) = \varepsilon \Lambda_r(t) v_r + O(\varepsilon^2)$. The eigenmode with the largest $\Lambda_r(t)$ is therefore the least restored and is the first one whose score component points away from the origin. Since the eigenvectors lie in $\mathcal{S}_{\text{pat}}$, this mode is a linear combination of the templates—the leading principal component of their covariance. This identifies only the *initial* unstable direction. Once x is no longer small, nonlinear posterior reweighting and tanh saturation determine which discrete template basin is selected. If the top eigenvalue is degenerate, the spinodal selects the whole top eigenspace rather than a unique direction. The spinodal time and the posterior recovery time need not coincide: the former is a threshold-free local curvature transition, whereas the latter depends on a prescribed error level and on pairwise competitors. Their common $\frac{1}{2} \log D$ scaling arises when the relevant covariance or separation eigenvalues are of order D .

6.6 Exact score on binary templates

In the binary setting $h_i^\mu = H\sigma_i^\mu$, the per-pattern denoiser at site j becomes

$$\hat{a}_j^\mu(x, t) = \tanh(\beta H \sigma_j^\mu + \lambda(t) x_j). \quad (95)$$

The posterior-averaged denoiser is then

$$\hat{a}_j(x, t) = \sum_{\mu=1}^M P(\mu | x, t) \tanh(\beta H \sigma_j^\mu + \lambda(t) x_j). \quad (96)$$

Pattern overlaps and magnetisation dynamics. A natural way to track selection through the score is via the *overlap* of the denoiser with each pattern. Define

$$q_\mu(x, t) := \frac{1}{D} \sum_{j=1}^D \sigma_j^\mu \hat{a}_j(x, t). \quad (97)$$

This measures how much the posterior-averaged denoiser resembles the template σ^μ . Before selection, several overlaps can be comparable. After selection of μ^* , one expects q_{μ^*} to be largest, while a competitor ν retains an overlap controlled by its template correlation $\rho_{\mu^*\nu}$; it is close to zero only for nearly orthogonal patterns. Substituting (96) into (97):

$$q_\mu(x, t) = \sum_{\nu=1}^M P(\nu | x, t) \underbrace{\frac{1}{D} \sum_{j=1}^D \sigma_j^\mu \tanh(\beta H \sigma_j^\nu + \lambda(t) x_j)}_{=: Q_{\mu\nu}(x, t)}.$$

The matrix $Q_{\mu\nu}(x, t)$ encodes the “cross-talk” between patterns as seen through the denoiser. Its diagonal entries $Q_{\mu\mu}$ measure how well pattern μ ’s own denoiser aligns with its template; its off-diagonal entries $Q_{\mu\nu}$ ($\mu \neq \nu$) measure the spurious alignment with competitor templates.

7 Two-level clustering

We now consider four binary templates arranged into two close pairs,

$$\mathcal{C}_1 = \{1, 2\}, \quad \mathcal{C}_2 = \{3, 4\}.$$

The cluster label $C = C(\mu) \in \{1, 2\}$ is a deterministic coarse-graining of the original latent index; it is not an additional physical variable. This section distinguishes two notions once again: posterior recovery of a cluster or component, and local curvature spinodals of the noised density.

7.1 Exact posteriors

The cluster posterior is

$$P(C = c \mid x, t) = \sum_{\mu \in \mathcal{C}_c} P(\mu \mid x, t) = \frac{\sum_{\mu \in \mathcal{C}_c} P_0(\mu) e^{S_\mu(x, t)}}{\sum_{\kappa=1}^4 P_0(\kappa) e^{S_\kappa(x, t)}}. \quad (98)$$

Define the outer log-odds

$$\ell_{\text{out}}(x, t) := \log \frac{P(C = 1 \mid x, t)}{P(C = 2 \mid x, t)}. \quad (99)$$

Because there are only two clusters,

$$\hat{C}(x, t) = \begin{cases} 1, & \ell_{\text{out}}(x, t) \geq 0, \\ 2, & \ell_{\text{out}}(x, t) < 0, \end{cases} \quad \kappa_{\text{out}}(x, t) = |\ell_{\text{out}}(x, t)|.$$

There is no additional minimum over competing clusters. Conditioning on $C = c$ gives

$$P(\mu \mid C = c, x, t) = \frac{P_0(\mu) e^{S_\mu(x, t)}}{\sum_{\nu \in \mathcal{C}_c} P_0(\nu) e^{S_\nu(x, t)}}, \quad \mu \in \mathcal{C}_c. \quad (100)$$

Since each cluster contains two templates, the two possible fine margins are

$$\kappa_{\text{in},1}(x, t) = |\ell_{12}(x, t)|, \quad \kappa_{\text{in},2}(x, t) = |\ell_{34}(x, t)|. \quad (101)$$

After a cluster has been selected, only the margin inside that cluster is relevant.

7.2 Two-scale geometry and a balanced realization

For binary templates,

$$\|m^\mu - m^\nu\|_2^2 = 4m_0^2 d_{\mu\nu}, \quad m_0 := \tanh(\beta H), \quad d_{\mu\nu} := d_H(\sigma^\mu, \sigma^\nu), \quad (102)$$

and, with $\rho_{\mu\nu} = D^{-1} \langle \sigma^\mu, \sigma^\nu \rangle$,

$$d_{\mu\nu} = \frac{D}{2} (1 - \rho_{\mu\nu}). \quad (103)$$

The posterior-recovery analysis only requires

$$d_{12} = d_{34} =: d_{\text{in}}, \quad d_{\text{out}} := \min_{\mu \in \mathcal{C}_1, \nu \in \mathcal{C}_2} d_{\mu\nu}, \quad d_{\text{out}} \gg d_{\text{in}}.$$

However, the curvature calculation at $x = 0$ also requires the balance condition (78). A particularly transparent realization satisfying both requirements is

$$\sigma^3 = -\sigma^1, \quad \sigma^4 = -\sigma^2, \quad d_H(\sigma^1, \sigma^2) = d_{\text{in}} =: k. \quad (104)$$

Then the two pairs are equally weighted and sitewise balanced, $\sum_{\mu=1}^4 \sigma^\mu = 0$, while

$$d_{12} = d_{34} = k, \quad d_{13} = d_{24} = D, \quad d_{14} = d_{23} = D - k, \quad d_{\text{out}} = D - k.$$

Thus $k \ll D$ gives the desired hierarchy without moving the high-noise stationary point away from the origin.

7.3 Posterior-recovery times

The outer transition should be derived at the cluster level rather than by selecting an arbitrary cross-cluster pair. Define the conditional mean-spin vectors

$$\bar{m}_c := \mathbb{E}[m^\mu \mid C = c] = \sum_{\mu \in \mathcal{C}_c} P_0(\mu \mid C = c) m^\mu.$$

Under the balanced construction (104), with $u = \sigma^1$ and $v = \sigma^2$,

$$\bar{m}_1 = \frac{m_0}{2}(u + v), \quad \bar{m}_2 = -\bar{m}_1, \quad \|\bar{m}_1\|_2^2 = m_0^2(D - k). \quad (105)$$

Expanding the exact cluster log-odds (99) at high noise gives

$$\ell_{\text{out}}(x, t) \approx \lambda(t) \langle x, \bar{m}_1 - \bar{m}_2 \rangle = 2\lambda(t) \langle x, \bar{m}_1 \rangle.$$

Under a planted observation from cluster 1, $\mathbb{E}[x \mid C = 1] = e^{-t} \bar{m}_1$. Neglecting the subleading signal contribution to the variance, the corresponding class-level SNR is therefore

$$\text{SNR}_{\text{out}}(t) \approx \frac{e^{-t}}{\sqrt{\Delta_t}} \|\bar{m}_1\|_2 = \frac{m_0 e^{-t} \sqrt{D - k}}{\sqrt{\Delta_t}}. \quad (106)$$

Thus the outer formula depends on the separation of the two *cluster means*, not on a notational choice of one representative from each cluster. Inside cluster $\mathcal{C}_1$, the two mean-spin vectors are $m_0 u$ and $m_0 v$. Since they have equal norm and Hamming distance k , the pairwise coefficient is

$$C_{12} = \frac{\langle m_0 u, m_0(u - v) \rangle}{\|m_0(u - v)\|_2} = m_0 \sqrt{k},$$

and the same holds inside $\mathcal{C}_2$. Consequently, the two high-noise planted-recovery times are

$$\boxed{\begin{aligned} t_{\text{out}}^{\text{rec}}(L_{\text{out}}) &\approx \frac{1}{2} \log \left(1 + \frac{m_0^2(D - k)}{L_{\text{out}}^2 T} \right), \\ t_{\text{in}}^{\text{rec}}(L_{\text{in}}) &\approx \frac{1}{2} \log \left(1 + \frac{m_0^2 k}{L_{\text{in}}^2 T} \right). \end{aligned}} \quad (107)$$

These are confidence-dependent recovery crossovers based on a high-noise Gaussian approximation. The exact cluster error and exact within-pair error should be measured in Monte Carlo. If the same SNR threshold L is used at both levels and both logarithmic arguments are large,

$$t_{\text{out}}^{\text{rec}} - t_{\text{in}}^{\text{rec}} \sim \frac{1}{2} \log \frac{D - k}{k}. \quad (108)$$

For $k = D^\alpha$ with $0 < \alpha < 1$, the scales are respectively $\frac{1}{2} \log D$ and $\frac{\alpha}{2} \log D$. If $k = O(1)$, the inner transition must be determined from the exact error curve (65), not from a CLT.

Numerical benchmark. For $D = 10^3$, $k = 10$, $m_0 = 0.9$, $L_{\text{out}} = L_{\text{in}} = 3$, and $T = 1$,

$$t_{\text{out}}^{\text{rec}} \approx \frac{1}{2} \log(90.1) \approx 2.25, \quad t_{\text{in}}^{\text{rec}} \approx \frac{1}{2} \log(1.9) \approx 0.32.$$

The corresponding OU variances are

$$\Delta_{t_{\text{out}}} \approx 1 - \frac{1}{90.1} \approx 0.9889, \quad \Delta_{t_{\text{in}}} \approx 1 - \frac{1}{1.9} \approx 0.4737.$$

Here $L = 3$ corresponds, under the Gaussian approximation, to the much stricter binary accuracy $\Phi(3) \approx 0.99865$. The Monte Carlo study below uses a 95% binary target, for which $L_{0.95} := \Phi^{-1}(0.95) \approx 1.64485$; its empirical crossing times are therefore expected to be larger. The large separation in forward time means that reverse generation first acquires the pair identity while the member identity remains uncertain.

7.4 Exact curvature spinodals for the balanced pair geometry

Under (104), write $u := \sigma^1$ and $v := \sigma^2$. The pattern kernel is

$$K = \frac{1}{4} \sum_{\mu=1}^4 \sigma^\mu \sigma^{\mu\top} = \frac{1}{2} (uu^\top + vv^\top).$$

Since $\langle u, v \rangle = D - 2k$, the vectors $u + v$ and $u - v$ are orthogonal eigenvectors of K :

$$K(u + v) = (D - k)(u + v), \quad K(u - v) = k(u - v).$$

Thus the nonzero eigenvalues are

$$\gamma_{\text{out}} = D - k, \quad \gamma_{\text{in}} = k.$$

The first mode represents the features shared by the two members of a pair and distinguishes the pair from its sign-reversed partner. The second mode represents the coordinates on which the members of a pair differ. Substitution into the exact mode formula (93) gives

$$\boxed{\begin{aligned} t_{\text{spin,out}} &= \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2(D - k)}{T} \right), \\ t_{\text{spin,in}} &= \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2 k}{T} \right). \end{aligned}} \quad (109)$$

These equalities are exact for the Hessian at the balanced origin. They do not contain a confidence threshold L , because a spinodal records the first loss of local concavity rather than a prescribed recovery probability. When $D \gg k \gg 1$ and the informative terms dominate,

$$t_{\text{spin,out}} - t_{\text{spin,in}} \sim \frac{1}{2} \log \frac{D - k}{k},$$

which is the same geometric separation as in (108). For the numerical parameters above,

$$t_{\text{spin,out}} \approx 3.34, \quad t_{\text{spin,in}} \approx 1.11.$$

These times are larger than the $L = 3$ recovery times because the spinodal marks the threshold-free onset of amplification, whereas a recovery criterion asks for enough accumulated evidence to achieve a specified error probability. The two should be compared rather than identified.

7.5 Intermediate denoising regime

Suppose $\mathcal{C}_1 = \{1, 2\}$ has high posterior mass but its members remain nearly tied. Then

$$\hat{a}_j(x, t) \approx \frac{1}{2} [\tanh(\beta H \sigma_j^1 + \lambda x_j) + \tanh(\beta H \sigma_j^2 + \lambda x_j)].$$

At consensus sites, $\sigma_j^1 = \sigma_j^2$, the two terms coincide and the common sign is already resolved. At disagreement sites, $\sigma_j^1 = -\sigma_j^2$, the two biases cancel at $x_j = 0$ and the remaining response is

$$\lambda(t)(1 - m_0^2)x_j + O((\lambda(t)x_j)^3).$$

Thus the denoiser can resolve the shared coarse features before the k coordinates that distinguish the two members.

7.6 Simulation-ready observables

At each fixed time t , the Monte Carlo study should estimate:

1. component recovery $\mathbb{P}(\hat{\mu} = \mu^*)$;
2. cluster recovery $\mathbb{P}(\hat{C} = C(\mu^*))$;
3. the posterior peak $\max_{\mu} P(\mu \mid x, t)$ and posterior entropy;

4. the outer and inner margins $|\ell_{\text{out}}|$ and $|\ell_{12}|$ or $|\ell_{34}|$;
5. the exact pairwise error curve $p_{\text{err}}(t; d)$, the exact moment ratio (61), and its approximations (66)–(68);
6. the eigenvalues of the exact Hessian $\nabla_x^2 \log P_t(0)$, to compare posterior-recovery crossovers with curvature spinodals.

This separates three questions that should not be conflated: posterior confidence, correct planted recovery, and local loss of concavity.

7.7 Two-Cluster Monte Carlo Simulation

We now test the two-stage recovery predictions by sampling from the exact balanced two-pair model. The experiment is performed independently at each fixed forward time t , so it tests the posterior formulas directly and does not introduce a time-discretization error from the reverse SDE. For each realization we draw

$$\mu^* \sim \text{Unif}\{1, 2, 3, 4\}, \quad a \sim P_0(\cdot \mid \mu^*), \quad x = e^{-t}a + \sqrt{\Delta_t}z, \quad z \sim \mathcal{N}(0, I_D),$$

with

$$\Delta_t = T(1 - e^{-2t}),$$

and evaluate the exact finite- D posterior $P(\mu \mid x, t)$.

Let $C^* := C(\mu^*)$ be the planted cluster. We define

$$\hat{C}(x, t) := \arg \max_{c \in \{1, 2\}} P(C = c \mid x, t),$$

and, conditional on the planted cluster,

$$\hat{\mu}_{\text{in}}(x, t) := \arg \max_{\mu \in \mathcal{C}_{C^*}} P(\mu \mid C = C^*, x, t).$$

The empirical outer, inner, and global recovery probabilities are

$$A_{\text{out}}(t) := \mathbb{P}(\hat{C} = C^*), \tag{110}$$

$$A_{\text{in}}(t) := \mathbb{P}(\hat{\mu}_{\text{in}} = \mu^*), \tag{111}$$

$$A_{\text{glob}}(t) := \mathbb{P}\left(\arg \max_{1 \leq \mu \leq 4} P(\mu \mid X, t) = \mu^*\right). \tag{112}$$

We use the target accuracy $1 - \varepsilon = 0.95$, corresponding under the one-sided Gaussian approximation to

$$L_{0.95} := \Phi^{-1}(0.95) \approx 1.64485. \tag{113}$$

The empirical recovery time is the largest forward time for which the isotonic fit of the corresponding accuracy remains at least 0.95. Wilson intervals quantify pointwise uncertainty and a joint bootstrap quantifies uncertainty in the two crossing times and in their difference.

Detailed benchmark. The principal run uses

$$D = 400, \quad k = 20, \quad m_0 = 0.9, \quad T = 1,$$

with 2500 independent samples at each of 52 time points and 300 bootstrap replications. The effective dimensions are

$$n_{\text{out}} = D - k = 380, \quad n_{\text{in}} = k = 20.$$

For a generic effective dimension n , the high-noise and first-order recovery predictions plotted below are

$$t^{\text{HN}}(L; n) = \frac{1}{2} \log \left(1 + \frac{m_0^2 n}{L^2 T} \right), \tag{114}$$

$$t^{(1)}(L; n) = \frac{1}{2} \log \left(1 + \frac{m_0^2 n - L^2(1 - m_0^2)}{L^2 T} \right), \tag{115}$$

where the second formula requires a positive numerator. The exact curvature spinodal associated with the same mode is

$$t_{\text{spin}}(n) = \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2 n}{T} \right). \quad (116)$$

For the benchmark parameters,

	t^{HN}	$t^{(1)}$	t_{spin}
outer	2.3714	2.3706	2.8667
inner	0.9721	0.9583	1.4279

while the empirical crossings are

$$\hat{t}_{\text{out}}^{\text{MC}} = 2.3985, \quad \hat{t}_{\text{in}}^{\text{MC}} = 0.9526,$$

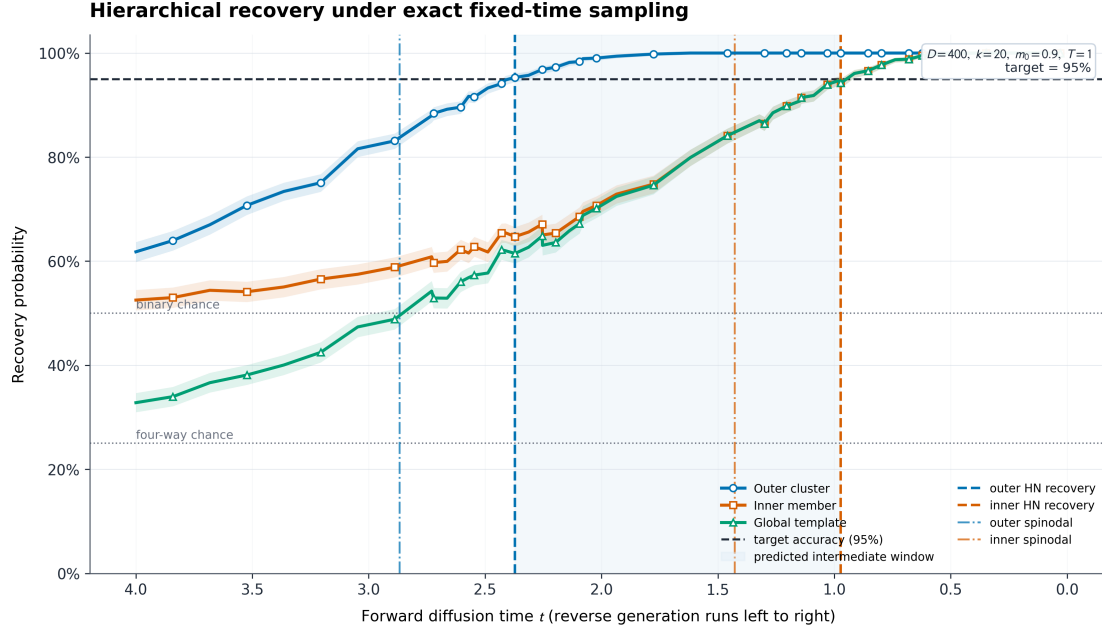
with bootstrap intervals

$$[2.3479, 2.4233], \quad [0.9350, 0.9891].$$

The empirical hierarchy gap is

$$\hat{t}_{\text{out}}^{\text{MC}} - \hat{t}_{\text{in}}^{\text{MC}} = 1.4405, \quad 95\% \text{ CI} = [1.3840, 1.4757]. \quad (117)$$

All 300 bootstrap realizations produced a positive gap.



Quantities plotted on the vertical axis.

$$y = A_{\text{out}}(t), \quad y = A_{\text{in}}(t), \quad y = A_{\text{glob}}(t),$$

with the three probabilities defined in (110)–(112). The horizontal target is

$$y = 0.95.$$

Vertical theoretical lines.

$$t_{\text{out}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{m_0^2(D-k)}{L_{0.95}^2 T} \right) = 2.3714,$$

$$t_{\text{in}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{m_0^2 k}{L_{0.95}^2 T} \right) = 0.9721.$$

The curvature lines are

$$t_{\text{spin,out}} = \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2(D-k)}{T} \right) = 2.8667,$$

$$t_{\text{spin,in}} = \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2 k}{T} \right) = 1.4279.$$

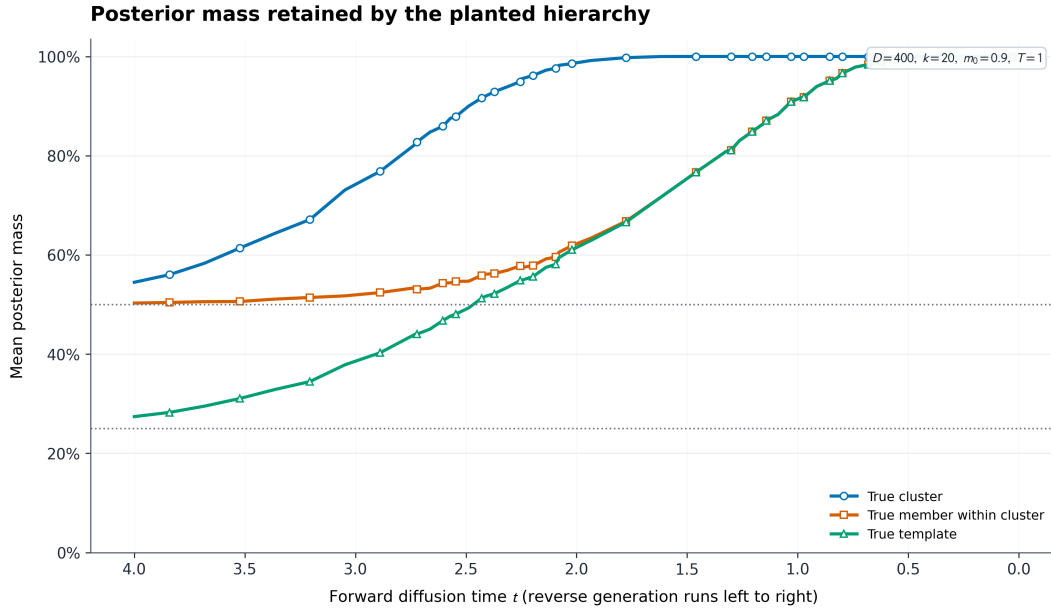
Numerical agreement. The high-noise crossing errors are 0.0270 and 0.0194, giving

$$\text{MSE}_t^{\text{HN}} = 5.54 \times 10^{-4}, \quad \text{RMSE}_t^{\text{HN}} = 0.0235.$$

The first-order crossing RMSE is 0.0201. At the high-noise predicted times the observed accuracies are 0.9532 and 0.9424; two-sided calibration tests against 0.95 give $p = 0.4913$ and $p = 0.0893$, and neither is rejected after Holm correction.

Figure 1: Exact fixed-time recovery probabilities. The time axis is reversed so that reverse generation proceeds from left to right. The plot exhibits a non-empty interval in which cluster recovery is already reliable but member recovery is not.

Recovery curves.



Formulas plotted.

$$\begin{aligned}
 y &= M_C(t) := \mathbb{E}[P(C^* \mid X, t)], \\
 y &= M_{\text{in}}(t) := \mathbb{E}[P(\mu^* \mid C = C^*, X, t)], \\
 y &= M_\mu(t) := \mathbb{E}[P(\mu^* \mid X, t)].
 \end{aligned}$$

At infinite noise, the reference values are

$$M_C(\infty) = \frac{1}{2}, \quad M_{\text{in}}(\infty) = \frac{1}{2}, \quad M_\mu(\infty) = \frac{1}{4}.$$

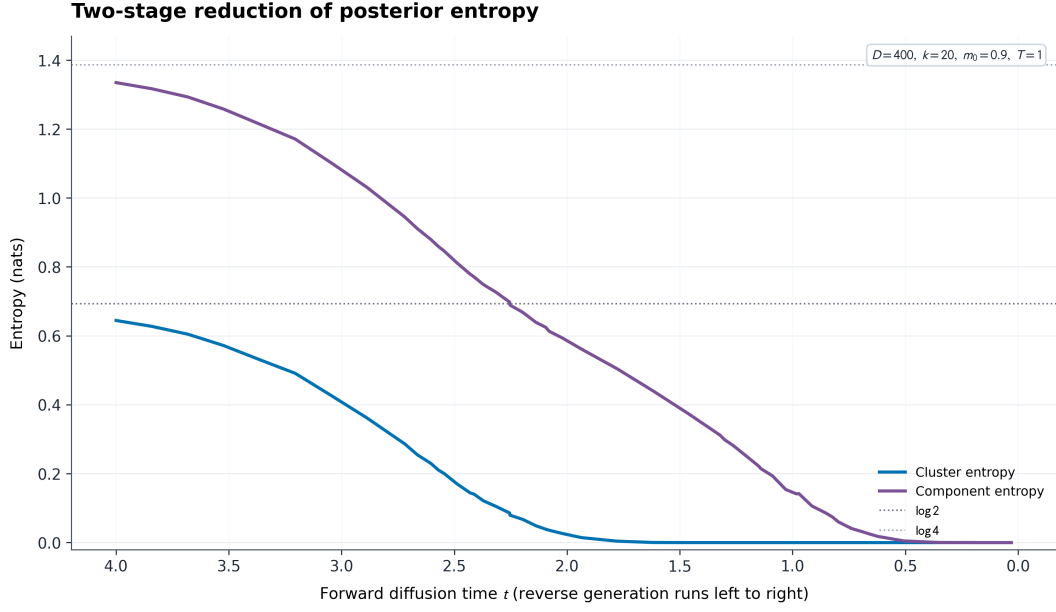
Representative numerical values.

t	$M_C(t)$	$M_{\text{in}}(t)$	$M_\mu(t)$
4.000	0.5448	0.5029	0.2740
2.196	0.9617	0.5787	0.5564
0.030	1.0000	1.0000	1.0000

At $t = 2.196$, the planted cluster already retains 96.17% of the posterior mass, whereas the conditional member mass is only 57.87%.

Figure 2: Posterior mass retained by the planted hierarchy. The separation between M_C and M_{in} is a threshold-free manifestation of the coarse-to-fine transition.

Posterior masses.



Formulas plotted.

$$y = H_C(t) := \mathbb{E} \left[- \sum_{c=1}^2 P(C = c \mid X, t) \log P(C = c \mid X, t) \right],$$

$$y = H_\mu(t) := \mathbb{E} \left[- \sum_{\mu=1}^4 P(\mu \mid X, t) \log P(\mu \mid X, t) \right].$$

The ideal sequence is

$$(H_C, H_\mu) : (\log 2, \log 4) \longrightarrow (0, \log 2) \longrightarrow (0, 0).$$

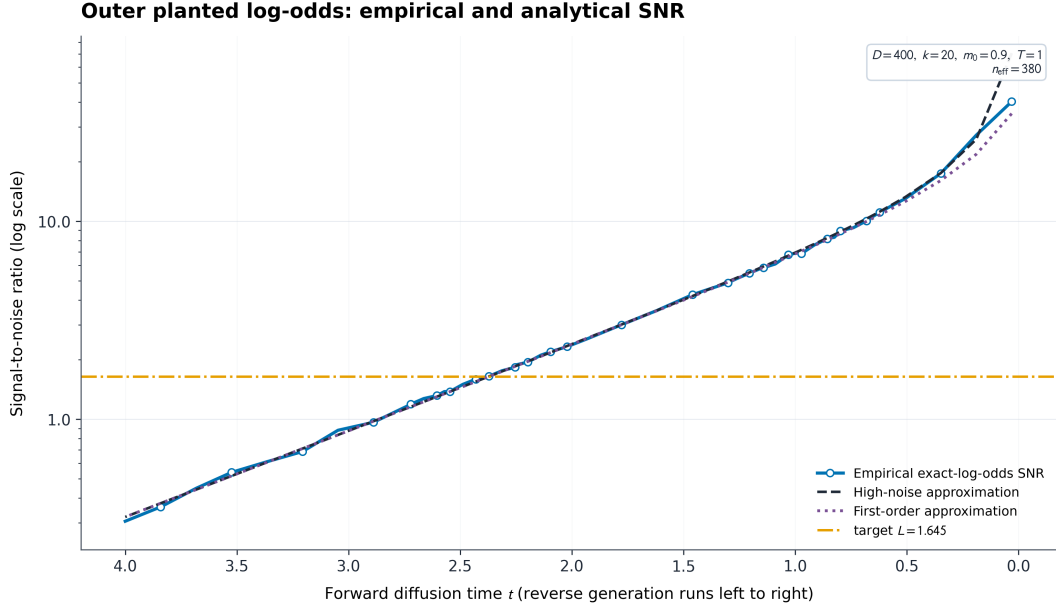
Representative numerical values.

t	$H_C(t)$	$H_\mu(t)$
4.000	0.6444	1.3346
2.196	0.0673	0.6681
0.030	≈ 0	1.17×10^{-12}

At $t = 2.196$, $H_C = 0.0673$ is already close to zero, while $H_\mu = 0.6681$ remains close to $\log 2 \approx 0.6931$, indicating one unresolved binary choice inside the selected cluster.

Figure 3: Expected cluster and component posterior entropies. The cluster entropy collapses first, while the component entropy remains near $\log 2$ in the intermediate regime.

Posterior entropies.



Empirical quantity plotted.

$$y = \widehat{\text{SNR}}_{\text{out}}(t) := \frac{\bar{\ell}_{\text{out}}(t)}{s_{\text{out}}(t)},$$

where

$$\ell_{\text{out}}^*(X, t) := \log \frac{P(C = C^* | X, t)}{P(C \neq C^* | X, t)}.$$

Analytical curves plotted. For $n_{\text{out}} = D - k = 380$,

$$y = \text{SNR}_{\text{out}}^{\text{HN}}(t) = \frac{m_0 \sqrt{D - k} e^{-t}}{\sqrt{T(1 - e^{-2t})}}$$

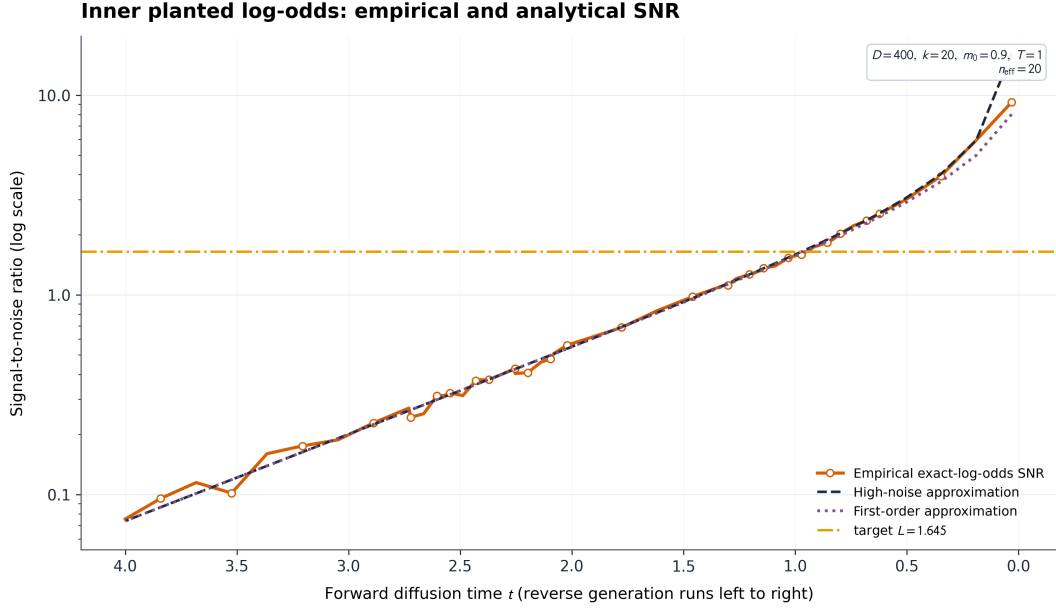
and

$$y = \text{SNR}_{\text{out}}^{(1)}(t) = \frac{m_0 \sqrt{D - k} e^{-t}}{\sqrt{T(1 - e^{-2t}) + e^{-2t}(1 - m_0^2)}}.$$

The horizontal line is $y = L_{0.95} = 1.64485$. In the transition window $|t - t_{\text{out}}^{\text{HN}}| \leq 0.2$, the high-noise and first-order RMSE values are 0.0260 and 0.0266. At $t = t_{\text{out}}^{\text{HN}} = 2.3714$, the empirical value is 1.6492.

Figure 4: Empirical and analytical outer-cluster SNR. Because $D - k = 380$ is large, the high-noise approximation is already highly accurate near the crossing.

Outer signal-to-noise ratio.



Empirical quantity plotted.

$$y = \widehat{\text{SNR}}_{\text{in}}(t) := \frac{\bar{\ell}_{\text{in}}(t)}{s_{\text{in}}(t)},$$

where $\bar{\mu}^*$ is the other member of the planted cluster and

$$\ell_{\text{in}}^*(X, t) := \log \frac{P(\mu^* | X, t)}{P(\bar{\mu}^* | X, t)}.$$

Analytical curves plotted. For $n_{\text{in}} = k = 20$,

$$y = \text{SNR}_{\text{in}}^{\text{HN}}(t) = \frac{m_0 \sqrt{k} e^{-t}}{\sqrt{T(1 - e^{-2t})}}$$

and

$$y = \text{SNR}_{\text{in}}^{(1)}(t) = \frac{m_0 \sqrt{k} e^{-t}}{\sqrt{T(1 - e^{-2t}) + e^{-2t}(1 - m_0^2)}}.$$

The horizontal line is $y = L_{0.95}$. In the transition window $|t - t_{\text{in}}^{\text{HN}}| \leq 0.2$, the high-noise approximation has

$$\text{RMSE} = 0.0380,$$

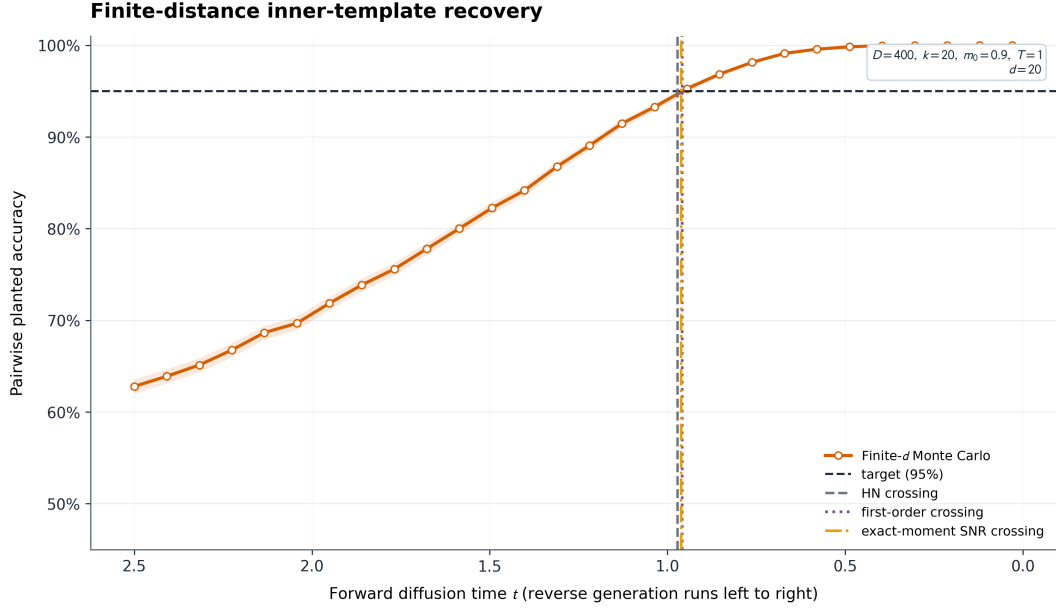
while retaining spin variance lowers it to

$$\text{RMSE} = 0.0225.$$

At $t = t_{\text{in}}^{\text{HN}} = 0.9721$, the empirical SNR is 1.5864 and the observed inner accuracy is 0.9424.

Figure 5: Empirical and analytical inner-template SNR. The finite- k spin variance is visible here, so the first-order approximation improves on the high-noise curve.

Inner signal-to-noise ratio.



At a disagreeing site, define

$$g_r(t) := 2 \operatorname{artanh}[m_0 \tanh(\lambda(t)X'_r)], \quad X'_r = e^{-t}A'_r + \eta_r,$$

with

$$\mathbb{P}(A'_r = 1) = \frac{1 + m_0}{2}, \quad \eta_r \sim \mathcal{N}(0, \Delta_t).$$

Formula plotted. For $d = k = 20$,

$$L_d(t) := \sum_{r=1}^d g_r(t),$$

$$y = A_d(t) = \mathbb{P}(L_d(t) > 0) + \frac{1}{2}\mathbb{P}(L_d(t) = 0).$$

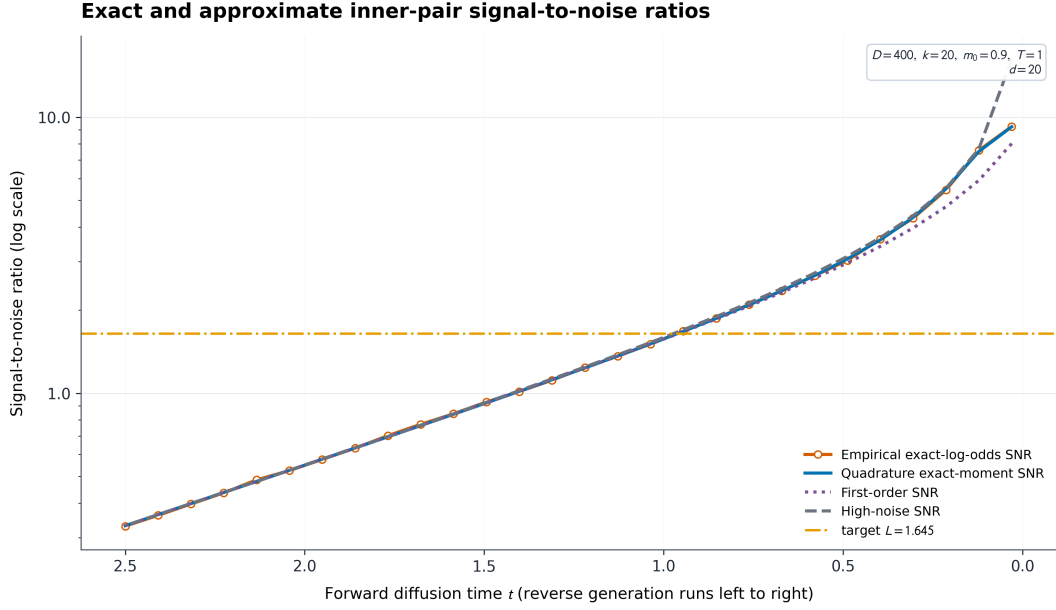
The empirical 95% crossing is 0.9571. The three vertical predictions are

$$t_d^{\text{HN}} = 0.9721, \quad t_d^{(1)} = 0.9583, \quad t_d^{\text{mom}} = 0.9613,$$

with absolute crossing errors 0.0150, 0.0012, and 0.0042. The exact clean-endpoint error is 5.94×10^{-9} .

Figure 6: Exact finite-distance inner recovery. The first-order and exact-moment crossing predictions closely track the Monte Carlo crossing.

Exact finite- k inner accuracy.



Exact-moment curve plotted.

$$\mu_g(t) := \mathbb{E}[g_r(t)], \quad \sigma_g^2(t) := \text{Var}(g_r(t)),$$

$$y = \text{SNR}_{\text{mom}}(t; d) = \sqrt{d} \frac{\mu_g(t)}{\sigma_g(t)}.$$

The plot also contains the empirical SNR of $L_d(t)$ and the high-noise and first-order formulas given above. In the selection window $0.55 \leq t \leq 1.35$, the RMSE relative to the empirical SNR is

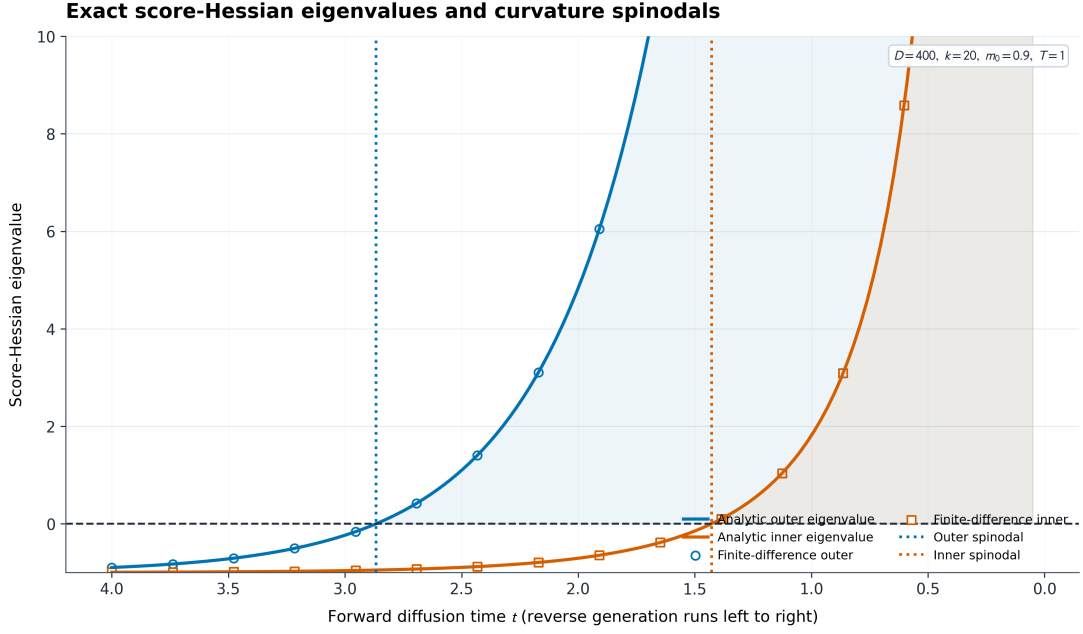
0.00388 for the exact-moment curve,

0.0235 for the first-order curve, 0.0294 for the high-noise curve.

If the SNRs are converted into Gaussian accuracy predictions $A(t) = \Phi(\text{SNR}(t))$, the all-grid RMSE values are 0.001584, 0.001573, and 0.002105, respectively.

Figure 7: Exact and approximate SNRs for the finite- k inner decision. The quadrature exact-moment curve gives the most accurate SNR description in the transition window.

Exact finite- k inner SNR.



Curves plotted. For a kernel eigenvalue γ ,

$$y = \Lambda_\gamma(t) = -\frac{1}{\Delta_t} + \frac{e^{-2t}}{\Delta_t^2} (1 - m_0^2 + m_0^2 \gamma), \quad \Delta_t = T(1 - e^{-2t}).$$

The two modes use

$$\gamma_{\text{out}} = D - k = 380, \quad \gamma_{\text{in}} = k = 20.$$

Solving $\Lambda_\gamma(t) = 0$ yields

$$t_{\text{spin}}(\gamma) = \frac{1}{2} \log \left(1 + \frac{1 - m_0^2 + m_0^2 \gamma}{T} \right),$$

so that

$$t_{\text{spin,out}} = 2.8667, \quad t_{\text{spin,in}} = 1.4279.$$

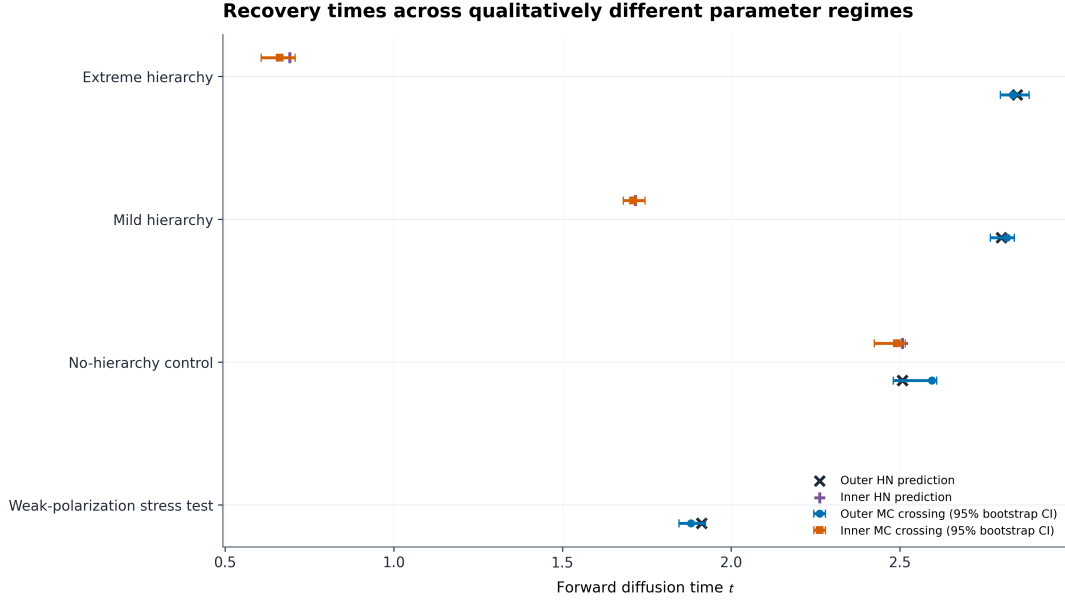
Central finite differences of the exact score have outer and inner RMSE 1.20×10^{-3} and 3.25×10^{-6} . The largest absolute errors are 4.80×10^{-3} and 1.30×10^{-5} ; the former occurs where $|\Lambda_{\text{out}}| \approx 1.20 \times 10^4$.

Figure 8: Exact score-Hessian eigenvalues and finite-difference checks. The spinodals occur at larger forward times than the 95% recovery crossings, separating the onset of local amplification from reliable classification.

Score-Hessian spinodals.

Parameter dependence across four regimes. For every scenario below, the theoretical recovery times displayed in the plots are obtained from the same pair of formulas

$$t_{\text{out}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{m_0^2(D-k)}{L_{0.95}^2 T} \right), \quad t_{\text{in}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{m_0^2 k}{L_{0.95}^2 T} \right). \quad (118)$$



Formula represented by the theoretical markers. The crosses and plus signs are the two values in (118); the circles and squares are the empirical isotonic crossings with bootstrap confidence intervals.

Across all four outer crossings,

$$\text{RMSE}_{\text{out}} = 0.0476.$$

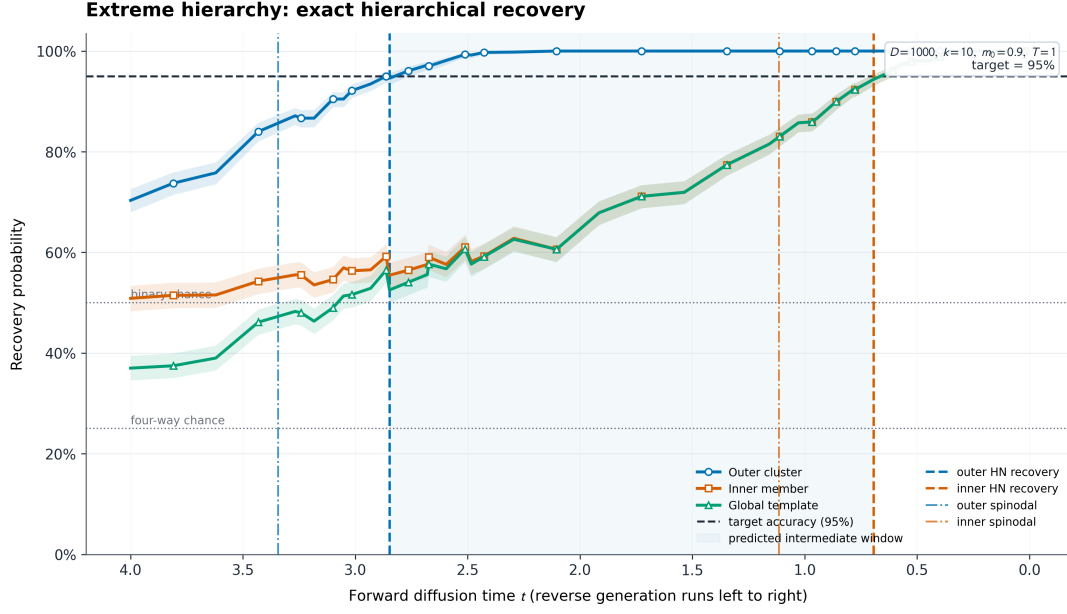
Across the three feasible inner crossings,

$$\text{RMSE}_{\text{in}} = 0.0203.$$

Pooling all seven feasible crossings gives

$$\text{MSE}_{\text{pooled}} = 1.47 \times 10^{-3}, \quad \text{RMSE}_{\text{pooled}} = 0.0384.$$

Figure 9: Empirical recovery crossings and high-noise predictions across four parameter regimes.



Here $D = 1000$, $k = 10$, $m_0 = 0.9$, and $T = 1$. Formula (118) gives

$$t_{\text{out}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.9^2(990)}{L_{0.95}^2} \right) = 2.8475,$$

$$t_{\text{in}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.9^2(10)}{L_{0.95}^2} \right) = 0.6924.$$

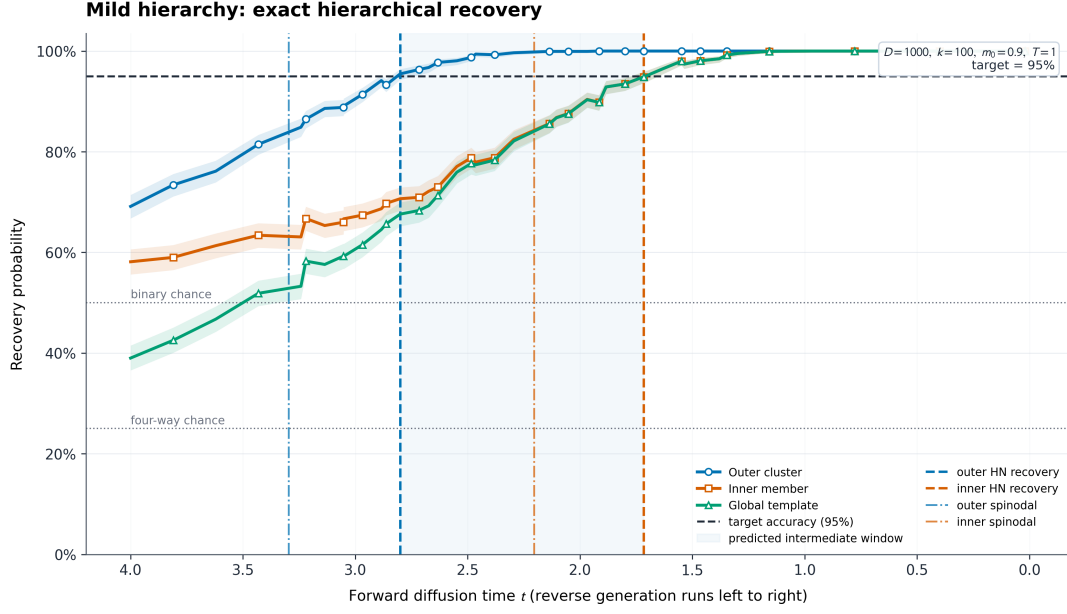
The empirical values are

$$\hat{t}_{\text{out}}^{\text{MC}} = 2.8343, \quad \hat{t}_{\text{in}}^{\text{MC}} = 0.6621,$$

with intervals $[2.7971, 2.8830]$ and $[0.6078, 0.7078]$. The absolute prediction errors are 0.0132 and 0.0303, and the empirical hierarchy gap is 2.1721.

Figure 10: Extreme hierarchy: a wide coarse-but-not-fine recovery window.

Extreme hierarchy.



Here $D = 1000$, $k = 100$, $m_0 = 0.9$, and $T = 1$. Formula (118) gives

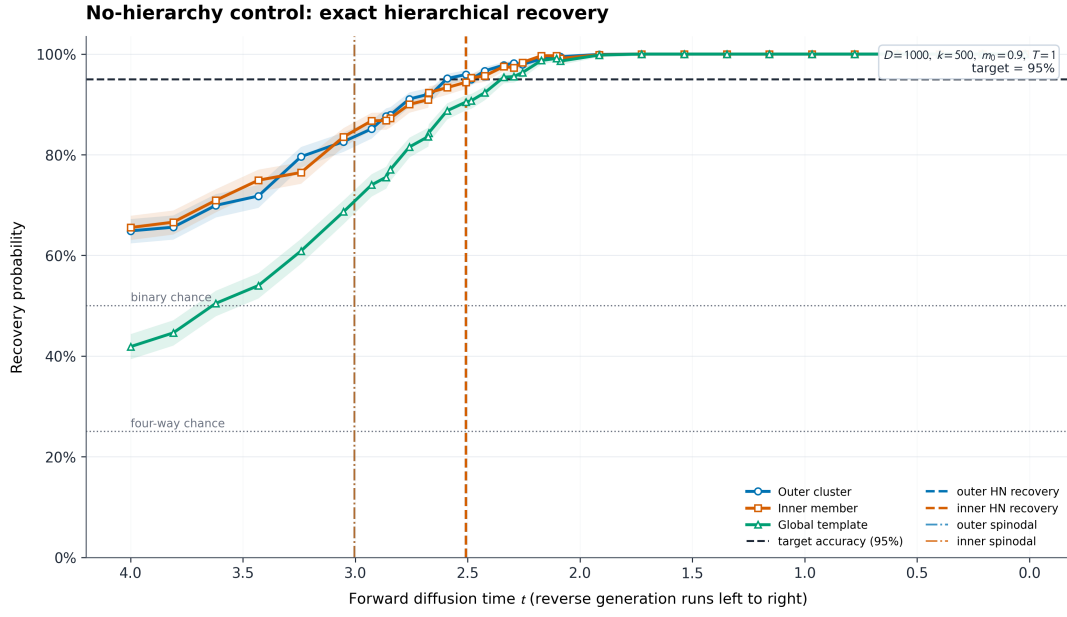
$$t_{\text{out}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.9^2(900)}{L_{0.95}^2} \right) = 2.8000,$$

$$t_{\text{in}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.9^2(100)}{L_{0.95}^2} \right) = 1.7160.$$

The empirical values are 2.8166 and 1.7090, with intervals $[2.7676, 2.8387]$ and $[1.6803, 1.7443]$. The absolute errors are 0.0166 and 0.0070, and the empirical gap is 1.1076.

Figure 11: Mild hierarchy: increasing k moves the fine crossing toward the coarse crossing and contracts the intermediate regime.

Mild hierarchy.



Here $D = 1000$, $k = 500$, $m_0 = 0.9$, and $T = 1$, so that

$$D - k = k = 500.$$

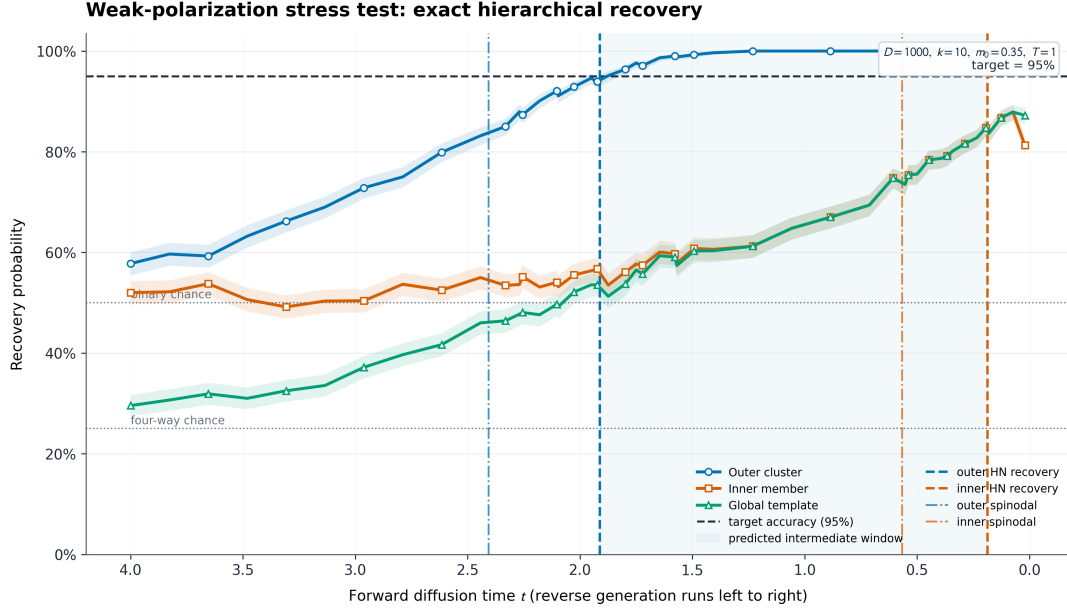
Hence formula (118) predicts the same time at both levels:

$$t_{\text{out}}^{\text{HN}} = t_{\text{in}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.9^2(500)}{L_{0.95}^2} \right) = 2.5076.$$

The empirical crossings are 2.5949 and 2.4911, with overlapping intervals $[2.4803, 2.6088]$ and $[2.4236, 2.5161]$. The empirical gap is only 0.1037, and its bootstrap positivity frequency is 0.96, rather than 1 as in the hierarchical cases.

Figure 12: No-hierarchy control: when $D - k = k$, the two leading recovery scales coincide.

No-hierarchy control.



Here $D = 1000$, $k = 10$, $m_0 = 0.35$, and $T = 1$. Formula (118) gives

$$t_{\text{out}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.35^2(990)}{L_{0.95}^2} \right) = 1.9124,$$

$$t_{\text{in}}^{\text{HN}} = \frac{1}{2} \log \left(1 + \frac{0.35^2(10)}{L_{0.95}^2} \right) = 0.1867.$$

The empirical outer crossing is 1.8808, with interval $[1.8456, 1.9235]$. The inner target is never reached: the maximum simulated inner accuracy is 0.8789. Indeed,

$$m_0^2 k = 1.225 < L_{0.95}^2 (1 - m_0^2) \approx 2.374,$$

so the first-order feasibility condition fails. At the clean endpoint,

$$\mathbb{P}(\text{Bin}(10, 0.675) > 5) + \frac{1}{2} \mathbb{P}(\text{Bin}(10, 0.675) = 5) \approx 0.8676.$$

Thus the formal high-noise root 0.1867 is not an attainable 95% recovery time.

Figure 13: Weak polarization: outer recovery survives aggregation over $D - k$ coordinates, but the finite- k inner decision cannot reach the chosen target.

Weak-polarization stress test.

Centered-RMS consistency check. Under the centered null approximation, the tested linear contrast is

$$Y(t) := \lambda(t) \langle X, \Delta m \rangle,$$

with theoretical variance

$$\text{Var}(Y(t)) = \frac{e^{-2t}}{\Delta_t} \|\Delta m\|_2^2.$$

At the outer and inner high-noise crossing times, the empirical-to-predicted variance ratios are 1.0136 and 1.0026. The corresponding chi-square variance-test p -values are 0.337 and 0.795. The empirical RMS values are 3.3124 and 3.2939, compared with the common prediction 3.2897.

Monte Carlo conclusion

Every displayed figure is tied to an explicit analytical quantity. For the baseline benchmark, the high-noise and first-order crossing RMSE values are 0.0235 and 0.0201. The exact finite- k SNR has transition-window RMSE 0.00388, and the score-Hessian finite-difference errors are negligible. Across the parameter suite, the pooled high-noise crossing RMSE is 0.0384 whenever the requested crossing exists.

The experiments support the predicted dependence on $D - k$, k , m_0 , and T , while also showing that a formal solution of the high-noise equation is not a recovery guarantee when the finite- k clean endpoint cannot attain the target accuracy. Recovery crossings and curvature spinodals remain distinct: the former impose a chosen confidence level, whereas the latter record the threshold-free loss of local concavity.

Symbol / expression	Meaning / role	Notable limits / behavior
$\mathcal{S}_D = \{\pm 1\}^D$	Discrete configuration space for spins a (or s).	(static)
$h^\mu \in \mathbb{R}^D$	Fixed template/pattern (index $\mu = 1, \dots, M$).	(fixed)
$P_0(a) \propto \sum_{\mu=1}^M e^{\beta \langle h^\mu, a \rangle}$	Static prior over configurations (mixture of Boltzmann factors).	(static)
$P_0(a \mid \mu) = \frac{e^{\beta \langle h^\mu, a \rangle}}{Z_\mu}$	Conditional spin law given pattern μ (product measure).	(static)
$\prod_{i=1}^D \frac{e^{\beta h_i^\mu a_i}}{2 \cosh(\beta h_i^\mu)}$		
$Z_\mu = \sum_a e^{\beta \langle h^\mu, a \rangle}$	Component partition function for pattern μ .	(static)
$\prod_{i=1}^D 2 \cosh(\beta h_i^\mu)$		
$P_0(\mu) = \frac{Z_\mu}{\sum_\nu Z_\nu}$	Static weight of component μ (auxiliary-pattern posterior under no observation).	If unique maximizer of $F_\mu^{(D)}$ has positive gap: $P_0(\mu^*) \rightarrow 1$ as $D \rightarrow \infty$.
$\frac{e^{DF_\mu^{(D)}}}{\sum_\nu e^{DF_\nu^{(D)}}}$		
$F_\mu^{(D)} = \frac{1}{D} \sum_{i=1}^D \log \cosh(\beta h_i^\mu)$	Component log-partition density controlling dominance among patterns.	Appears in $Z = 2^D \sum_\mu e^{DF_\mu^{(D)}}$; Laplace dominates as $D \rightarrow \infty$ (fixed M).
$m_i^\mu := \tanh(\beta h_i^\mu)$, $m^\mu = (m_i^\mu)_{i=1}^D$	Polarization/mean spin under $P_0(\cdot \mid \mu)$: $\mathbb{E}[a_i \mid \mu] = m_i^\mu$.	(static)
$\Delta m^{\mu\nu} := m^\mu - m^\nu$	Difference of polarizations between two patterns (direction that separates μ vs ν).	(static)
$x \mid a, t \sim \mathcal{N}(e^{-t}a, \Delta_t I)$	OU forward diffusion/noising channel.	$t \rightarrow 0$: $x \approx a$ (low noise); $t \rightarrow \infty$: $x \Rightarrow \mathcal{N}(0, TI)$ (high noise).
$\Delta_t = T(1 - e^{-2t})$	OU noise variance at time t .	$t \rightarrow 0$: $\Delta_t \rightarrow 0$; $t \rightarrow \infty$: $\Delta_t \rightarrow T$.
$\lambda(t) := \frac{e^{-t}}{\Delta_t} = \frac{1}{2T \sinh t}$	Effective signal-to-noise gain appearing in the posterior evidence.	$t \rightarrow \infty$: $\lambda(t) \sim e^{-t}/T \rightarrow 0$; $t \downarrow 0$: $\lambda(t) \sim (2Tt)^{-1} \rightarrow \infty$.
$P_t(x \mid \mu) = \sum_a P_0(a \mid \mu) p(x \mid a, t)$	Pattern-conditional likelihood of observing x at time t .	$t \rightarrow \infty$: becomes nearly μ -independent (no information); $t \rightarrow 0$: concentrates around $x \approx a$ with $a \sim P_0(\cdot \mid \mu)$.
$S_\mu(x, t) = \sum_{i=1}^D [\log \cosh(\beta h_i^\mu \lambda(t)x_i) - \log \cosh(\beta h_i^\mu)]$	Scalar dynamic log-evidence for pattern μ + (the μ -dependent part of the component log-likelihood).	$t \rightarrow \infty$: $\lambda \rightarrow 0$ so $S_\mu(x, t) \rightarrow 0$; small- λ expansion: $S_\mu \approx \lambda \langle x, m^\mu \rangle + \frac{\lambda^2}{2} \sum_i x_i^2 (1 - (m_i^\mu)^2) + \dots$.
$\frac{P(\mu \mid x, t)}{P_0(\mu) e^{S_\mu(x, t)}}$	Posterior over patterns given x at diffusion time t .	$t \rightarrow \infty$: $P(\mu \mid x, t) \rightarrow P_0(\mu)$ (data uninformative); $t \downarrow 0$: reduces to $P_0(\mu \mid a)$; concentration additionally requires sufficiently separated components.
$\frac{P(\mu \mid x, t)}{P(\nu \mid x, t)} = \log \frac{P_0(\mu)}{P_0(\nu)} + (S_\mu - S_\nu)$	Log-posterior odds (pairwise preference of pattern μ over ν).	At high noise: $\ell_{\mu\nu} \approx \log \frac{P_0(\mu)}{P_0(\nu)} + \lambda \langle x, \Delta m^{\mu\nu} \rangle$; typical magnitude grows as t decreases (via RMS).
$\kappa_{\text{post}}(x, t) = \min_{\nu \neq \hat{\mu}} \ell_{\hat{\mu}\nu}(x, t)$, $\hat{\mu} = \arg \max_\mu P(\mu \mid x, t)$	Posterior-winner margin; controls concentration through $P(\hat{\mu} \mid x, t) \geq [1 + (M - 1)e^{-\kappa_{\text{post}}}]^{-1}$.	$t \rightarrow \infty$: small for balanced priors; as t decreases its typical scale can grow, although individual trajectories need not be monotone.

Table 1: Notation summary: static vs diffusion quantities, their roles, and their notable limiting behavior as $t \rightarrow \infty$ (high noise) and $t \downarrow 0$ (data scale).